EXTRA PROBLEMS · 2ND MIDTERM · PROBLEMS

2nd Midterm: Exam-Style Problems

Exam-style problems on nonlinear systems, Fourier series and PDEs, worth 5 points each, like the problems of the second partial, written for this portal and solved step by step. They are practice material, not official exercises; each one was checked with SymPy.

7 exam-style problems on nonlinear systems, Fourier series and the heat, wave and Laplace equations, worth 5 points each. The second partials of 2019 to 2025 are two problems in 1 to 2 hours; the Exam simulator draws one of each block.

2nd Midterm · Unit 4 · Nonlinear systems and stability · 5 points · 45 min · Conceptual / fast

P08 — Two competing species with a competition parameter b: equilibria, Jacobians, the threshold b = 4/3 between coexistence and competitive exclusion

Two species share a habitat. Their populations $x(t) \geq 0$, $y(t) \geq 0$ (in thousands of individuals, time in years) obey

$$x' = x(6 - 2x - y), \qquad y' = y(4 - y - bx),$$

where $b > 0$ is a constant.

(a) Interpret each term of the model. Find all the equilibria as functions of b and the values of b for which there is an equilibrium with both species present ($x > 0$, $y > 0$).

(b) Compute the Jacobian matrix and classify every equilibrium in the closed first quadrant as b varies. Identify the threshold value of b that separates coexistence from competitive exclusion.

(c) Take $b = 1$. Draw (describe) the nullclines, the eigen-directions at each equilibrium and the phase portrait in the first quadrant. What happens in the long run to a population with $x(0) > 0$, $y(0) > 0$?

(d) Take $b = 2$. Answer the same questions. Which species survives?

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknowns $x(t), y(t)$; independent variable $t \geq 0$; the right-hand side is polynomial, hence C^1, so solutions exist and are unique (Picard). Autonomous, **nonlinear** (quadratic terms), planar.
- The axes are invariant: $x = 0 \Rightarrow x' = 0$, $y = 0 \Rightarrow y' = 0$. By uniqueness no orbit crosses them, so the closed first quadrant is invariant (populations stay non-negative).
- Parameter: $b > 0$, the harm done to y by each unit of x.

Coefficient	Meaning
6, 4	intrinsic growth rates of x, y (year^{-1})
$-2x^2$, $-y^2$	intraspecific competition; carrying capacities $K_x = 3$, $K_y = 4$
$-xy$ in x'	harm to x from encounters with y
$-bxy$ in y'	harm to y from encounters with x

Phase 2. Canonical formulation

$\mathbf{x}' = \mathbf{F}(\mathbf{x})$ with $\mathbf{F}(x, y) = \big(x(6 - 2x - y),\ y(4 - y - bx)\big)^T$. Near an equilibrium $\mathbf{x}_0$, $\mathbf{u} = \mathbf{x} - \mathbf{x}_0$ obeys $\mathbf{u}' \approx D\mathbf{F}(\mathbf{x}_0)\,\mathbf{u}$; when the equilibrium is hyperbolic (no eigenvalue with zero real part) the Hartman-Grobman theorem (Robinson Sec. 32.4.3) says the nonlinear portrait near it is the linear one.

Part (a) · 1 point

Interpretation, equilibria and existence of coexistence

Step 1. Interpretation

Each species grows logistically when alone ($y = 0$: $x' = 2x(3 - x)$, capacity 3; $x = 0$: $y' = y(4 - y)$, capacity 4). The cross terms $-xy$ and $-bxy$ are both negative: each species reduces the growth of the other, so the model is one of **competition** (Robinson Sec. 33.1). The larger b, the stronger the pressure of x on y.

Step 2. Equilibria

$x' = 0$ when $x = 0$ or $2x + y = 6$; $y' = 0$ when $y = 0$ or $y + bx = 4$. The four combinations:

- $x = 0, y = 0$: $(0, 0)$.
- $x = 0, y + bx = 4$: $(0, 4)$.
- $2x + y = 6, y = 0$: $(3, 0)$.
- $2x + y = 6$ and $bx + y = 4$: subtracting, $(2 - b)x = 2$, so for $b \neq 2$

$$x^* = \frac{2}{2 - b}, \qquad y^* = 6 - 2x^* = \frac{6(2 - b) - 4}{2 - b} = \frac{8 - 6b}{2 - b}.$$

Step 3. Positivity

$x^* > 0 \iff b < 2$. Then $2 - b > 0$ and $y^* > 0 \iff 8 - 6b > 0 \iff b < \frac{4}{3}$. So the coexistence point is in the open quadrant exactly for $0 < b < \frac{4}{3}$. At $b = \frac{4}{3}$: $x^* = \frac{2}{2/3} = 3$, $y^* = 0$, it coincides with $(3, 0)$. For $\frac{4}{3} < b < 2$ it lies in $y < 0$ (no biological meaning); for $b = 2$ the two lines are parallel and there is none.

KEY RESULTS

- Equilibria: $(0, 0)$, $(3, 0)$, $(0, 4)$ and, for $b \neq 2$, $\left(\dfrac{2}{2 - b}, \dfrac{8 - 6b}{2 - b} \right)$.
- The coexistence point lies in $x > 0, y > 0$ if and only if $0 < b < \frac{4}{3}$; at $b = \frac{4}{3}$ it merges with $(3, 0)$.

Part (b) · 2 points

Jacobian and classification as b varies

Step 1. Jacobian

$\dfrac{\partial}{\partial x}\left[6x - 2x^2 - xy\right] = 6 - 4x - y$, $\dfrac{\partial}{\partial y}\left[6x - 2x^2 - xy\right] = -x$, $\dfrac{\partial}{\partial x}\left[4y - y^2 - bxy\right] = -by$, $\dfrac{\partial}{\partial y}\left[4y - y^2 - bxy\right] = 4 - 2y - bx$.

Step 2. Origin

$D\mathbf{F}(0, 0) = \begin{pmatrix} 6 & 0 \\ 0 & 4 \end{pmatrix}$: two positive eigenvalues, **unstable node**, eigenvectors along the axes. Orbits leave tangent to the slow direction, the y axis ($4 < 6$).

Step 3. The point (0, 4)

$D\mathbf{F}(0, 4) = \begin{pmatrix} 6 - 4 & 0 \\ -4b & 4 - 8 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ -4b & -4 \end{pmatrix}$. Lower triangular: eigenvalues 2 and -4, a **saddle** for every b. Eigenvector of -4: $(0, 1)$, the y axis is the stable manifold. Eigenvector of 2: $(D\mathbf{F} - 2I)\mathbf{v} = \mathbf{0}$ gives row 2 $-4bv_1 - 6v_2 = 0$, so $\mathbf{v} = (3, -2b)^T$: the unstable manifold enters the quadrant towards the lower right. Meaning: a few individuals of x introduced into the y-only state grow (their per capita rate there is $6 - 4 = 2 > 0$).

Step 4. The point (3, 0)

$D\mathbf{F}(3, 0) = \begin{pmatrix} 6 - 12 & -3 \\ 0 & 4 - 3b \end{pmatrix} = \begin{pmatrix} -6 & -3 \\ 0 & 4 - 3b \end{pmatrix}$. Upper triangular: eigenvalues -6 (along the x axis) and $4 - 3b$, which is the per capita growth rate of a few y individuals in the x-only state.

- $b < \frac{4}{3}$: $4 - 3b > 0$, **saddle** (y can invade).
- $b > \frac{4}{3}$: $4 - 3b < 0$, **stable node** (y cannot invade).
- $b = \frac{4}{3}$: one zero eigenvalue, non-hyperbolic: the linearisation decides nothing there (Robinson Sec. 32.4.3). This is where the coexistence point crosses $(3, 0)$.

Step 5. Coexistence point

At (x^*, y^*): $6 - 4x^* - y^* = (6 - 2x^* - y^*) - 2x^* = -2x^*$ and $4 - 2y^* - bx^* = (4 - y^* - bx^*) - y^* = -y^*$, so

$$D\mathbf{F}(x^*, y^*) = \begin{pmatrix} -2x^* & -x^* \\ -by^* & -y^* \end{pmatrix}.$$

- $\mathrm{tr} = -(2x^* + y^*) = -6$ (because $2x^* + y^* = 6$ on the nullcline).
- $\det = 2x^*y^* - bx^*y^* = (2-b)x^*y^* = (2-b)\dfrac{2}{2-b}\dfrac{8-6b}{2-b} = \dfrac{2(8-6b)}{2-b}$, positive for $0 < b < \frac{4}{3}$.
- Discriminant: $\mathrm{tr}^2 - 4\det = (-2x^* - y^*)^2 - 4(2x^*y^* - bx^*y^*) = (2x^* - y^*)^2 + 4bx^*y^* > 0$.

Real eigenvalues, both negative ($\mathrm{tr} < 0$, $\det > 0$): **stable node** for every $0 < b < \frac{4}{3}$. (The off-diagonal entries have the same sign, which always forbids a focus in a competition model.)

Step 6. Summary and threshold

b	$(0,0)$	$(0,4)$	$(3,0)$	**coexistence**
$0 < b < \frac{4}{3}$	unstable node	saddle	saddle	stable node in the quadrant
$b = \frac{4}{3}$	unstable node	saddle	non-hyperbolic	merged with $(3,0)$
$b > \frac{4}{3}$	unstable node	saddle	stable node	outside the quadrant or none

The threshold is $b_c = \frac{4}{3}$, the value at which y stops being able to invade the x-only state: $4 - 3b_c = 0$, i.e. $b_c = r_y/K_x$ (growth rate of y over the carrying capacity of x).

KEY RESULTS

- $D\mathbf{F} = \begin{pmatrix} 6 - 4x - y & -x \\ -by & 4 - 2y - bx \end{pmatrix}$.
- $(0,0)$: $\mathrm{diag}(6, 4)$, unstable node for all b.
- $(0,4)$: $\begin{pmatrix} 2 & 0 \\ -4b & -4 \end{pmatrix}$, eigenvalues $2, -4$: saddle for all b.
- $(3,0)$: $\begin{pmatrix} -6 & -3 \\ 0 & 4 - 3b \end{pmatrix}$: saddle for $b < \frac{4}{3}$, stable node for $b > \frac{4}{3}$, non-hyperbolic at $b = \frac{4}{3}$.
- Coexistence ($0 < b < \frac{4}{3}$): $\begin{pmatrix} -2x^* & -x^* \\ -by^* & -y^* \end{pmatrix}$, $\mathrm{tr} = -6$, $\det = \dfrac{2(8-6b)}{2-b} > 0$, discriminant > 0: stable node.
- Threshold $b_c = \frac{4}{3}$.

Part (c) · 1 point

Phase portrait and long-time behaviour for b = 1

Step 1. Eigen-data at (2, 2)

$x^* = \frac{2}{1} = 2, y^* = \frac{2}{1} = 2$. $D\mathbf{F} = \begin{pmatrix} -4 & -2 \\ -2 & -2 \end{pmatrix}$, $p(\lambda) = (-4-\lambda)(-2-\lambda) - 4 = \lambda^2 + 6\lambda + 4$, $\lambda = -3 \pm \sqrt{5}$. Row 1 of $(D\mathbf{F} - \lambda I)\mathbf{v} = \mathbf{0}$: $(-4-\lambda)v_1 - 2v_2 = 0$, $v_2 = -\frac{4+\lambda}{2}v_1$. For $\lambda_s = -3 + \sqrt{5}$: $v_2 = -\frac{1+\sqrt{5}}{2}v_1$, $\mathbf{v}_s = (2, -1-\sqrt{5})$; for $\lambda_f = -3 - \sqrt{5}$: $v_2 = \frac{\sqrt{5}-1}{2}v_1$, $\mathbf{v}_f = (2, \sqrt{5}-1)$. Orbits reach $(2,2)$ tangent to $\mathbf{v}_s$ (slope $-\frac{1+\sqrt{5}}{2} \approx -1.62$), except the two on the fast line.

Step 2. Saddles

At $(3,0)$, $b = 1$: $\begin{pmatrix} -6 & -3 \\ 0 & 1 \end{pmatrix}$; for $\lambda = 1$, row 1: $-7v_1 - 3v_2 = 0$, $\mathbf{v} = (3, -7)$. Its branch pointing into the quadrant, along $(-3, 7)$, leaves $(3,0)$ upwards to the left and runs to $(2,2)$. At $(0,4)$: $\begin{pmatrix} 2 & 0 \\ -4 & -4 \end{pmatrix}$, unstable direction $(3, -2)$: its branch in the quadrant leaves $(0,4)$ to the lower right and runs to $(2,2)$. The axes are the stable manifolds of the two saddles.

Step 3. Nullclines and regions

L_x: $2x + y = 6$ (where $x' = 0$, vertical arrows), from $(0,6)$ to $(3,0)$. L_y: $x + y = 4$ (where $y' = 0$, horizontal arrows), from $(0,4)$ to $(4,0)$. They cross at $(2,2)$.

- Above both lines: $x' < 0, y' < 0$ (towards the lower left).
- Below both lines: $x' > 0, y' > 0$ (towards the upper right).
- Between them for $x < 2$ (e.g. $(1, 3.5)$): $x' > 0, y' < 0$.
- Between them for $x > 2$ (e.g. $(3, 0.5)$): $x' < 0, y' > 0$.

Step 4. Trapping regions and the long run

Take the triangle $R = \{x + y > 4,\ 2x + y < 6,\ x < 2\}$. On its side on L_x, $x' = 0$ and $y' = y(4 - x - y) = y(x - 2) < 0$ (there $y = 6 - 2x$, so $4 - x - y = x - 2$): the field points down, into R. On its side on L_y, $y' = 0$ and $x' = x(6 - 2x - y) = x(2 - x) > 0$: the field points right, into R. So R is positively invariant; inside it x increases and y decreases, both bounded, so they converge, and the limit must be an equilibrium: $(2,2)$. The other triangle is treated the same way. An orbit starting above or below both lines moves monotonically (lower left or upper right) until it either enters one of the triangles or converges directly to $(2,2)$. Hence every orbit with $x(0) > 0, y(0) > 0$ tends to $(2,2)$: the two species **coexist** at 2000 individuals each.

KEY RESULTS

- Coexistence point $(2,2)$; $D\mathbf{F} = \begin{pmatrix} -4 & -2 \\ -2 & -2 \end{pmatrix}$, $\lambda = -3 \pm \sqrt{5} \approx -0.764,\ -5.236$.
- Slow eigenvector $(2,\ -1 - \sqrt{5})$, fast eigenvector $(2,\ \sqrt{5} - 1)$.
- Saddles: $(3,0)$ with unstable direction $(3, -7)$ (eigenvalue 1); $(0,4)$ with unstable direction $(3, -2)$ (eigenvalue 2).
- Every orbit with $x(0) > 0, y(0) > 0$ tends to $(2,2)$: **coexistence**.

Part (d) · 1 point

Phase portrait and long-time behaviour for b = 2

Step 1. Equilibria and linearisation

$2x + y = 6$ and $2x + y = 4$ never meet, so only $(0,0)$, $(3,0)$, $(0,4)$ remain. At $(3,0)$, $4 - 3b = -2$: eigenvalues -6 and -2, a **stable node**; for $\lambda = -2$, row 1 of $(D\mathbf{F} + 2I)\mathbf{v} = \mathbf{0}$ is $-4v_1 - 3v_2 = 0$, so $\mathbf{v} = (3, -4)$, the slow direction to which orbits are tangent on arrival. At $(0,4)$, $\begin{pmatrix} 2 & 0 \\ -8 & -4 \end{pmatrix}$: a saddle, stable manifold the y axis, unstable direction from row 2 $-8v_1 - 6v_2 = 0$, $\mathbf{v} = (3, -4)$.

Step 2. Regions

- $2x + y > 6$: $x' < 0$ and $y' < 0$ (since also $2x + y > 4$).
- $4 < 2x + y < 6$: $x' > 0, y' < 0$ (e.g. $(1.5, 2)$): $x' = 1.5 \cdot 1 > 0, y' = 2 \cdot (-1) < 0$).
- $2x + y < 4$: $x' > 0, y' > 0$.

On $2x + y = 6$ the field is vertical and points down, on $2x + y = 4$ it is horizontal and points right: both point into the strip, which is positively invariant. In the strip x increases and y decreases, both bounded, so the orbit converges to an equilibrium in the closure of the strip with $x > 0$: $(3,0)$. Orbits outside the strip move monotonically into it or straight to $(3,0)$.

Step 3. Long run

Every orbit with $x(0) > 0$ (and any $y(0) \geq 0$) tends to $(3,0)$: species y becomes extinct and x reaches its carrying capacity. Only the orbits on the y axis ($x(0) = 0$) go to $(0,4)$. The unstable branch of the saddle $(0,4)$ runs to $(3,0)$, leaving along $(3,-4)$.

KEY RESULTS

- Equilibria $(0,0)$, $(3,0)$, $(0,4)$ only; nullclines $2x + y = 6$ and $2x + y = 4$, parallel.
- $(3,0)$: $\begin{pmatrix} -6 & -3 \\ 0 & -2 \end{pmatrix}$, stable node, slow eigenvector $(3,-4)$.
- $(0,4)$: $\begin{pmatrix} 2 & 0 \\ -8 & -4 \end{pmatrix}$, saddle, unstable eigenvector $(3,-4)$.
- Every orbit with $x(0) > 0$ tends to $(3,0)$: y dies out, **competitive exclusion**.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $b \to 0^+$: $x^* \to 1$, $y^* \to 4$; y no longer feels x, and x settles at $6 - 2x - 4 = 0$. $b \to \frac{4}{3}^-$: $(x^*, y^*) \to (3,0)$ and $\det D\mathbf{F} \to 0$, the exchange of stability with $(3,0)$ (a transcritical bifurcation, **extension**: a bifurcation where two equilibria cross and swap their stability). For every $b > \frac{4}{3}$ the outcome is that of (d).
- **Checks:** $\mathbf{F} = \mathbf{0}$ at each equilibrium; triangular Jacobians read directly; $\operatorname{tr} D\mathbf{F}(x^*, y^*) = -6$ for every b; at $b = 1$, $\lambda_s + \lambda_f = -6$ and $\lambda_s \lambda_f = 9 - 5 = 4 = \det$. Units: each bracket has units of year^{-1} when the coefficients carry year^{-1} and (thousand individuals)$^{-1}$ year^{-1}.

- **Common mistakes in the exam:** (1) keeping the coexistence point for $\frac{4}{3} < b < 2$ although $y^* < 0$; (2) using the full Jacobian formula at (x^*, y^*) without the simplification $6 - 2x^* - y^* = 0$, which hides that $\mathrm{tr} = -6$; (3) concluding stability at $b = \frac{4}{3}$ from the linearisation (zero eigenvalue); (4) predicting the long-time outcome from local stability alone, without the nullcline and trapping-region argument.

2nd Midterm · Unit 4 · Nonlinear systems and stability · 5 points · 45 min · Standard exam level

P09 — Conservative double-well oscillator x'' = 4x - x^3: linearisation, first integral, true centres, the figure-eight separatrix and the orbit types by energy

A particle of unit mass moves on a line under the force $4x - x^3$:

$$x'' = 4x - x^3.$$

(a) Write the equation as a first-order system for (x, y) with $y = x'$, and find the potential $V(x)$ with $x'' = -V'(x)$ and $V(0) = 0$. Find the equilibria and classify them by linearisation. Which conclusions can be transferred to the nonlinear system?

(b) Prove, by differentiating along solutions, that $E(x, y) = \frac{1}{2}y^2 + V(x)$ is constant on every orbit. Use it to show that the equilibria that are linear centres are true (nonlinear) centres, and give the sense of rotation around them.

(c) Find the explicit equation of the separatrix, the level set of E through the saddle, and its extent in x and y. Verify that $x(t) = 2\sqrt{2}\,\operatorname{sech}(2t)$ solves the equation with $x(0) = 2\sqrt{2}$, $x'(0) = 0$, and explain why this solution never reaches the saddle in finite time.

(d) Determine the ranges of E for which the orbits are: equilibria, closed orbits inside one well, the separatrix, closed orbits around all three equilibria. Give the turning points as functions of E and describe the phase portrait.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $x(t)$, $t \in \mathbb{R}$. Second-order, autonomous, **nonlinear** (cubic term), no damping. The right-hand side is C^1, so solutions exist and are unique (Picard).
- The force does not depend on x': the system is conservative (Robinson Sec. 34.1), and the energy will organise the whole portrait.
- Symmetries: the force is odd, so V is even and the portrait is symmetric under $x \mapsto -x$; replacing (y, t) by $(-y, -t)$ leaves the equation unchanged, so the portrait is symmetric under $y \mapsto -y$ with arrows reversed.

Phase 2. Canonical formulation

$$x' = y, \qquad y' = 4x - x^3 = -V'(x), \qquad V(x) = \frac{x^4}{4} - 2x^2.$$

Part (a) · 1.5 points

System, potential, equilibria and linearisation

Step 1. Potential

$$V(x) = -\int_0^x (4s - s^3)\,ds = -\Big[2s^2 - \frac{s^4}{4}\Big]_0^x = -\Big(2x^2 - \frac{x^4}{4}\Big) + 0 = \frac{x^4}{4} - 2x^2,$$

using Barrow's rule. Check: $V'(x) = x^3 - 4x = -(4x - x^3)$. It is a double well: V has a local maximum $V(0) = 0$ and two minima $V(\pm 2) = 4 - 8 = -4$.

Step 2. Equilibria

$y = 0$ and $4x - x^3 = x(2 - x)(2 + x) = 0$: $(0, 0)$, $(2, 0)$, $(-2, 0)$, the critical points of V.

Step 3. Linearisation

$D\mathbf{F} = \begin{pmatrix} 0 & 1 \\ 4 - 3x^2 & 0 \end{pmatrix}$, with $V''(x) = 3x^2 - 4$.

- At $(0, 0)$: $\begin{pmatrix} 0 & 1 \\ 4 & 0 \end{pmatrix}$, $\det(D\mathbf{F} - \lambda I) = \lambda^2 - 4$, $\lambda = \pm 2$. For $\lambda = 2$, row 1: $-2v_1 + v_2 = 0$, $\mathbf{v} = (1, 2)$; for $\lambda = -2$: $2v_1 + v_2 = 0$, $\mathbf{v} = (1, -2)$. Real eigenvalues of opposite sign: a **saddle**.
- At $(\pm 2, 0)$: $4 - 12 = -8$, $\begin{pmatrix} 0 & 1 \\ -8 & 0 \end{pmatrix}$, $\lambda^2 + 8 = 0$, $\lambda = \pm 2\sqrt{2}\,i$: a **linear centre**, with small-oscillation period $2\pi/(2\sqrt{2}) = \pi/\sqrt{2}$.

Step 4. What transfers

The saddle is hyperbolic, so by the Hartman-Grobman theorem (Robinson Sec. 32.4.3) the nonlinear system has a saddle at the origin with stable and unstable curves tangent to $(1,-2)$ and $(1,2)$. The centres are not hyperbolic (zero real part): a small nonlinear term could turn them into weak foci, so the linearisation alone decides nothing. Part (b) settles it.

KEY RESULTS

- $V = \dfrac{x^4}{4} - 2x^2$, $V(0)=0$, $V(\pm 2) = -4$.
- Equilibria $(0,0)$, $(2,0)$, $(-2,0)$.
- $(0,0)$: $\begin{pmatrix} 0 & 1 \\ 4 & 0 \end{pmatrix}$, $\lambda = \pm 2$, eigenvectors $(1,2)$ (unstable), $(1,-2)$ (stable): saddle, also for the nonlinear system.
- $(\pm 2,0)$: $\begin{pmatrix} 0 & 1 \\ -8 & 0 \end{pmatrix}$, $\lambda = \pm 2\sqrt{2}\,i$: linear centres, non-hyperbolic, linearisation inconclusive.

Part (b) · 1.5 points

First integral and true centres

Step 1. Conservation

Along a solution, by the chain rule,

$$\frac{d}{dt}E(x(t),y(t)) = \frac{\partial E}{\partial x}x' + \frac{\partial E}{\partial y}y' = V'(x)\,y + y\,\big(-V'(x)\big) = 0.$$

Explicitly: $(x^3-4x)y + y(4x-x^3) = 0$. So E is a first integral and every orbit lies on one level set $\frac{1}{2}y^2 + \frac{x^4}{4} - 2x^2 = E_0$.

Step 2. Shape of E near the centres

$E_x = V'(x)$, $E_y = y$ vanish at $(2,0)$, and the Hessian there is $\begin{pmatrix} V''(2) & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 8 & 0 \\ 0 & 1 \end{pmatrix}$, positive definite: E has a strict local minimum $E(2,0) = -4$. Near it, $E \approx -4 + 4(x-2)^2 + \frac{1}{2}y^2$, so for c slightly above -4 the level set $E = c$ is a small closed oval around $(2,0)$.

Step 3. Orbits on those level sets are periodic

More precisely, for $-4 < c < 0$ the right-well part of $\{E = c\}$ is $y = \pm\sqrt{2(c - V(x))}$ for $x \in [x_-, x_+]$, the interval where $V \le c$ (found in (d)): one closed curve, symmetric in y, meeting $y = 0$ only at the turning points. It contains no equilibrium ($V'(x_\pm) \neq 0$), so the speed never vanishes on it and the solution goes round it in a finite time and repeats: every nearby orbit is closed. The equilibria $(\pm 2,0)$ are therefore **true centres** of the nonlinear system (Robinson Sec. 34.1), by symmetry also $(-2,0)$.

Step 4. Sense of rotation

At $(2,1)$, above the centre, $(x',y') = (1,\ 4\cdot 2 - 8) = (1,0)$: the motion is to the right on top, so the orbits are travelled **clockwise**, as for every conservative system written with $y = x'$ (x increases where $y > 0$).

KEY RESULTS

- $\dfrac{dE}{dt} = V'(x)\,y + y\,(-V'(x)) = 0.$
- E has a strict local minimum -4 at $(\pm 2, 0)$ (Hessian $\operatorname{diag}(8,1)$): level sets $E = c$, $-4 < c < 0$, are closed curves around each minimum, so the orbits are periodic: **true centres**.
- Rotation clockwise (at $(2,1)$: $x' = 1 > 0$).

Part (c) · 1 point

Separatrix and the homoclinic solution

Step 1. Level set through the saddle

$E(0,0) = 0$. Then $\frac{1}{2}y^2 + \frac{x^4}{4} - 2x^2 = 0$ gives

$$y^2 = 4x^2 - \frac{x^4}{2} = \frac{x^2(8-x^2)}{2}, \qquad y = \pm x\sqrt{\frac{8-x^2}{2}}.$$

Real only for $x^2 \le 8$, i.e. $|x| \le 2\sqrt{2}$, where $V(\pm 2\sqrt{2}) = \frac{64}{4} - 16 = 0$. Near $x = 0$, $y \approx \pm x\sqrt{4} = \pm 2x$: the separatrix leaves the origin along the eigenvectors $(1, \pm 2)$ found in (a). It is a figure eight: one loop in $x > 0$ around $(2,0)$, its mirror image in $x < 0$.

Step 2. Height of the loops

$g(x) = \frac{x^2(8-x^2)}{2} = 4x^2 - \frac{x^4}{2}$, $g'(x) = 8x - 2x^3 = 2x(4-x^2) = 0$ at $x = \pm 2$, with $g(2) = 16 - 8 = 8$. So $|y| \le 2\sqrt{2}$, reached above and below the centres.

Step 3. Verification of the homoclinic solution

Let $x = 2\sqrt{2}\operatorname{sech} u$ with $u = 2t$, $\frac{du}{dt} = 2$. By the chain rule

$$x' = 2\sqrt{2}\,(-\operatorname{sech} u \tanh u)\cdot 2 = -4\sqrt{2}\operatorname{sech} u \tanh u.$$

Product rule, with $\frac{d}{dt}\tanh u = 2\operatorname{sech}^2 u$:

$$x'' = -4\sqrt{2}\big[(-2\operatorname{sech} u\tanh u)\tanh u + \operatorname{sech} u\,(2\operatorname{sech}^2 u)\big] = 8\sqrt{2}\operatorname{sech} u\tanh^2 u - 8\sqrt{2}\operatorname{sech}^3 u.$$

With $\tanh^2 u = 1 - \operatorname{sech}^2 u$: $x'' = 8\sqrt{2}\operatorname{sech} u - 16\sqrt{2}\operatorname{sech}^3 u$. On the other side, $4x - x^3 = 8\sqrt{2}\operatorname{sech} u - (2\sqrt{2})^3\operatorname{sech}^3 u = 8\sqrt{2}\operatorname{sech} u - 16\sqrt{2}\operatorname{sech}^3 u$. They agree. Initial data: $\operatorname{sech} 0 = 1$, $\tanh 0 = 0$, so $x(0) = 2\sqrt{2}$ and $x'(0) = 0$; $E = 0 + V(2\sqrt{2}) = 0$, the solution is on the separatrix.

Step 4. Infinite time

$\operatorname{sech} 2t > 0$ for every real t and $\operatorname{sech} 2t \to 0$ only as $t \to \pm\infty$: the solution leaves the origin at $t = -\infty$ along $(1, 2)$, reaches the turning point $x = 2\sqrt{2}$ at $t = 0$ and returns to the origin at $t = +\infty$ along $(1, -2)$. It is a **homoclinic orbit**. It cannot reach $(0, 0)$ in finite time, because $(0, 0)$ is itself a solution and uniqueness forbids two solutions to meet. Its speed is largest when $\operatorname{sech}^2 u = \frac{1}{2}$, with $|x'| = 4\sqrt{2} \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = 2\sqrt{2}$, as in Step 2.

KEY RESULTS

- Separatrix $E = 0$: $y^2 = \dfrac{x^2(8 - x^2)}{2}$, i.e. $y = \pm x\sqrt{\dfrac{8 - x^2}{2}}$, for $|x| \le 2\sqrt{2}$: a figure eight through the origin.
- Tangents at the origin $y = \pm 2x$ (the saddle eigenvectors); maximum $|y| = 2\sqrt{2}$ at $x = \pm 2$.
- $x(t) = 2\sqrt{2}\operatorname{sech} 2t$: $x'' = 8\sqrt{2}\operatorname{sech} 2t - 16\sqrt{2}\operatorname{sech}^3 2t = 4x - x^3$; $x \to 0$ only as $t \to \pm\infty$.

Part (d) · 1 point

Energy ranges, turning points and portrait

Step 1. Admissible energies

$E = \frac{1}{2}y^2 + V(x) \ge V(x) \ge -4$. So $E < -4$ gives an empty level set, and $E = -4$ forces $y = 0$, $V(x) = -4$: the two centres.

Step 2. Turning points

At a turning point $y = 0$ and $V(x) = E$: $\frac{x^4}{4} - 2x^2 - E = 0$, i.e. $X^2 - 8X - 4E = 0$ with $X = x^2$. The quadratic formula gives

$$X = \frac{8 \pm \sqrt{64 + 16E}}{2} = 4 \pm 2\sqrt{4 + E}.$$

- For $-4 < E < 0$: $0 < 2\sqrt{4 + E} < 4$, so both roots are positive and $x^2 \in [\,4 - 2\sqrt{4 + E},\ 4 + 2\sqrt{4 + E}\,]$ gives two disjoint intervals of x, one in each well: two closed orbits (one around each centre).

- For $E > 0$: $4 - 2\sqrt{4+E} < 0$ is rejected; the only turning points are $x = \pm\sqrt{4+2\sqrt{4+E}}$, and the level set is one closed curve surrounding all three equilibria, crossing $x = 0$ where $\frac{1}{2}y^2 = E$.

Step 3. Portrait

Inside each loop of the figure eight there is a family of nested clockwise ovals around $(\pm 2, 0)$, shrinking to the centre as $E \to -4^+$ and growing to the loop as $E \to 0^-$ (their period tends to infinity there). Outside the figure eight there are large clockwise closed curves, peanut-shaped (pinched at $x = 0$) for E slightly above 0, and close to the quartic curves $\frac{1}{2}y^2 + \frac{x^4}{4} = E$ for large E. Every orbit except the equilibria and the separatrix is periodic; there are no attractors, as expected without damping.

KEY RESULTS

- $E < -4$: no orbit; $E = -4$: the centres $(\pm 2, 0)$.
- $-4 < E < 0$: two closed orbits, one per well, between $x^2 = 4 - 2\sqrt{4+E}$ and $x^2 = 4 + 2\sqrt{4+E}$ (e.g. $E = -3$: $\sqrt{2} \le |x| \le \sqrt{6}$).
- $E = 0$: the saddle and the two homoclinic loops.
- $E > 0$: one closed orbit around all three equilibria, turning points $x = \pm\sqrt{4+2\sqrt{4+E}}$, crossing $x = 0$ with $y = \pm\sqrt{2E}$ (e.g. $E = 5$: $x = \pm\sqrt{10}$, $y = \pm\sqrt{10}$).

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $E \to -4^+$: small ellipses $4(x-2)^2 + \frac{1}{2}y^2 = E + 4$, period $\to \pi/\sqrt{2}$ (the linear value). $E \to 0$: the period grows without bound (the orbit lingers near the saddle). $E \to \infty$: $V \approx x^4/4$ dominates and the amplitude grows like $(4E)^{1/4}$.
- **Checks:** $dE/dt = 0$ identically; the separatrix leaves the origin with slopes ± 2, matching the saddle eigenvectors; the homoclinic solution has $E = 0$ and maximum speed $2\sqrt{2}$, matching the height of the loop. Dimensions, if x is in metres and t in seconds: the coefficients 4 and 1 carry s^{-2} and $\mathrm{m}^{-2}\mathrm{s}^{-2}$, and E is an energy per unit mass, $\mathrm{m}^2\mathrm{s}^{-2}$.
- **Common mistakes in the exam:** (1) stating that the linear centres are centres of the nonlinear system "by Hartman-Grobman": that theorem does not apply to non-hyperbolic points, the first integral is needed; (2) a sign error in V (taking $V' = 4x - x^3$), which turns minima into maxima and swaps saddles and centres; (3) forgetting the root $X = 4 - 2\sqrt{4+E}$ and describing one orbit instead of two for $-4 < E < 0$; (4) drawing the separatrix as crossing itself transversally at angles other than those of the eigenvectors $(1, \pm 2)$.

2nd Midterm · Unit 5 · Fourier series and convergence · 5 points · 45 min · Standard exam level

P10 — Full Fourier series on (-pi, pi) of the function equal to 0 on the left half and x^2 on the right half: coefficients, Dirichlet values and two numerical series

Let f be defined on $(-\pi, \pi)$ by

$$f(x) = \begin{cases} 0, & -\pi < x < 0, \\ x^2, & 0 \le x < \pi, \end{cases}$$

and extended to $\mathbb{R}$ as a 2π-periodic function.

(a) Describe the graph of the periodic extension on $[-3\pi, 3\pi]$: where is it continuous, where does it jump and by how much? Is f piecewise smooth?

(b) Compute the Fourier series $\dfrac{a_0}{2} + \displaystyle\sum_{n \ge 1} \big(a_n \cos nx + b_n \sin nx\big)$ of f.

(c) Using the Dirichlet theorem, give the sum of the series at $x = 0$, $x = \pi$, $x = -\pi$, $x = \pi/2$ and $x = -\pi/2$.

(d) Deduce the sums of $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$ and $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$, and check that the two results are consistent with each other.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknowns: the coefficients a_0, a_n, b_n ($n \ge 1$). Independent variable $x \in \mathbb{R}$, period 2π ($L = \pi$).
- f is bounded, and on each of $(-\pi, 0)$ and $(0, \pi)$ it is a polynomial: f and f' are piecewise continuous, so f is piecewise smooth and the Dirichlet theorem applies.
- No symmetry: f is neither even nor odd, so both a_n and b_n are needed. Since $f = 0$ on $(-\pi, 0)$, every integral reduces to $(0, \pi)$.

Phase 2. Canonical formulation

Euler formulas on $(-\pi, \pi)$:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx \ (n \ge 0), \qquad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx \ (n \ge 1).$$

Dirichlet theorem: for a piecewise smooth 2π-periodic f, the series converges at every x to $\frac{1}{2}\big[f(x^+) + f(x^-)\big]$.

Part (a) · 1 point

Periodic extension, continuity and jumps

Step 1. One period

On $(-\pi, 0)$ the graph is the segment $y = 0$; on $[0, \pi)$ it is the arc of $y = x^2$, rising from $(0, 0)$ to height π^2 as $x \to \pi^-$.

Step 2. Continuity at the origin

$f(0^-) = 0$, $f(0^+) = 0^2 = 0$: continuous. The one-sided derivatives are 0 (left) and $2 \cdot 0 = 0$ (right): the graph even has no corner at $x = 0$; only f'' jumps there (from 0 to 2).

Step 3. Jumps at the odd multiples of pi

The extension satisfies $f(x + 2\pi) = f(x)$. At $x = \pi$: $f(\pi^-) = \lim_{x \to \pi^-} x^2 = \pi^2$, while $f(\pi^+) = f(-\pi^+) = 0$. So the extension jumps down by π^2 at $x = \pm\pi, \pm 3\pi$. The graph on $[-3\pi, 3\pi]$ is three copies of the period: flat on $(-3\pi, -2\pi)$, arc on $(-2\pi, -\pi)$, flat on $(-\pi, 0)$, arc on $(0, \pi)$, flat on $(\pi, 2\pi)$, arc on $(2\pi, 3\pi)$, with a vertical drop from π^2 to 0 at each odd multiple of π.

KEY RESULTS

- On each period: zero on $(-\pi, 0)$, the parabola arc x^2 from 0 up to π^2 on $(0, \pi)$.
- Continuous at $x = 2k\pi$ ($f(0^-) = f(0^+) = 0$, and even $f'(0^\pm) = 0$).
- Jumps at $x = (2k+1)\pi$: $f(\pi^-) = \pi^2$, $f(\pi^+) = 0$, jump of size $-\pi^2$.
- Piecewise smooth: yes.

Part (b) · 2 points

Fourier coefficients

Step 1. Mean value

$$a_0 = \frac{1}{\pi}\int_0^{\pi} x^2\,dx = \frac{1}{\pi}\Big[\frac{x^3}{3}\Big]_0^{\pi} = \frac{1}{\pi}\cdot\frac{\pi^3}{3} = \frac{\pi^2}{3}.$$

Step 2. Two auxiliary integrals ($n \geq 1$)

By parts with $u = x$, $dv = \sin nx\,dx$, $du = dx$, $v = -\frac{\cos nx}{n}$:

$$\int x \sin nx\,dx = -\frac{x\cos nx}{n} + \frac{1}{n}\int \cos nx\,dx = -\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}.$$

By parts with $u = x$, $dv = \cos nx\,dx$, $du = dx$, $v = \frac{\sin nx}{n}$:

$$\int x \cos nx\,dx = \frac{x\sin nx}{n} - \frac{1}{n}\int \sin nx\,dx = \frac{x\sin nx}{n} + \frac{\cos nx}{n^2}.$$

Step 3. Cosine coefficients

By parts with $u = x^2$, $dv = \cos nx\,dx$, $du = 2x\,dx$, $v = \frac{\sin nx}{n}$, and then Step 2:

$$\int x^2 \cos nx\,dx = \frac{x^2\sin nx}{n} - \frac{2}{n}\int x\sin nx\,dx = \frac{x^2\sin nx}{n} + \frac{2x\cos nx}{n^2} - \frac{2\sin nx}{n^3}.$$

Barrow's rule on $[0,\pi]$, with $\sin n\pi = \sin 0 = 0$ and $\cos n\pi = (-1)^n$:

$$\int_0^{\pi} x^2\cos nx\,dx = \left[0 + \frac{2\pi(-1)^n}{n^2} - 0\right] - \left[0 + 0 - 0\right] = \frac{2\pi(-1)^n}{n^2}, \qquad a_n = \frac{2(-1)^n}{n^2}.$$

Step 4. Sine coefficients

By parts with $u = x^2$, $dv = \sin nx\,dx$, $du = 2x\,dx$, $v = -\frac{\cos nx}{n}$, and then Step 2:

$$\int x^2 \sin nx\,dx = -\frac{x^2\cos nx}{n} + \frac{2}{n}\int x\cos nx\,dx = -\frac{x^2\cos nx}{n} + \frac{2x\sin nx}{n^2} + \frac{2\cos nx}{n^3}.$$

Barrow's rule on $[0,\pi]$:

$$\int_0^{\pi} x^2 \sin nx\, dx = \Big[-\frac{\pi^2(-1)^n}{n} + 0 + \frac{2(-1)^n}{n^3} \Big] - \Big[0 + 0 + \frac{2}{n^3} \Big] = -\frac{\pi^2(-1)^n}{n} + \frac{2\big((-1)^n - 1\big)}{n^3}.$$

Dividing by π:

$$b_n = \frac{(-1)^{n+1}\pi}{n} + \frac{2\big((-1)^n - 1\big)}{\pi n^3} = \begin{cases} \dfrac{\pi}{n} - \dfrac{4}{\pi n^3}, & n \text{ odd}, \\ -\dfrac{\pi}{n}, & n \text{ even}. \end{cases}$$

Step 5. The series

$$f(x) \sim \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \left[\frac{2(-1)^n}{n^2} \cos nx + \left(\frac{(-1)^{n+1}\pi}{n} + \frac{2\big((-1)^n - 1\big)}{\pi n^3} \right) \sin nx \right].$$

First terms: $\frac{\pi^2}{6} - 2\cos x + \left(\pi - \frac{4}{\pi}\right)\sin x + \frac{1}{2}\cos 2x - \frac{\pi}{2}\sin 2x - \frac{2}{9}\cos 3x + \dots$

KEY RESULTS

- $a_0 = \dfrac{\pi^2}{3}$ (so the constant term is $\dfrac{\pi^2}{6}$).
- $a_n = \dfrac{2(-1)^n}{n^2}$: $a_1 = -2$, $a_2 = \frac{1}{2}$, $a_3 = -\frac{2}{9}$.
- $b_n = \dfrac{(-1)^{n+1}\pi}{n} + \dfrac{2\big((-1)^n - 1\big)}{\pi n^3}$: $b_n = \dfrac{\pi}{n} - \dfrac{4}{\pi n^3}$ for n odd, $b_n = -\dfrac{\pi}{n}$ for n even.

Part (c) · 1 point

Values of the series by the Dirichlet theorem

Step 1. Points of continuity

$x = 0$: f is continuous there, sum $f(0) = 0$. $x = \pi/2$: inside $(0, \pi)$, sum $(\pi/2)^2 = \pi^2/4$. $x = -\pi/2$: inside $(-\pi, 0)$, sum 0.

Step 2. The endpoints are jump points

For the periodic extension, $x = \pi$ and $x = -\pi$ are the same kind of point: $f(\pi^-) = \pi^2$, $f(\pi^+) = f(-\pi^+) = 0$. The series converges to

$$\frac{f(\pi^+)+f(\pi^-)}{2}=\frac{0+\pi^2}{2}=\frac{\pi^2}{2},$$

the same value at $x=-\pi$ (each term is 2π-periodic, so the series takes equal values at $\pm\pi$). It is not $f(\pi)$ of either formula: the series does not "choose" a side.

KEY RESULTS

- $x=0$: 0. $x=\pm\pi$: $\dfrac{\pi^2}{2}$. $x=\pi/2$: $\dfrac{\pi^2}{4}$. $x=-\pi/2$: 0.

Part (d) · 1 point

Two numerical series

Step 1. Evaluation at x = 0

$\cos 0=1$, $\sin 0=0$, and by (c) the sum is 0:

$$0=\frac{\pi^2}{6}+\sum_{n=1}^{\infty}\frac{2(-1)^n}{n^2}\implies\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}=-\frac{\pi^2}{12}\implies\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2}=1-\frac{1}{4}+\frac{1}{9}-\cdots=\frac{\pi^2}{12}.$$

Step 2. Evaluation at x = pi

$\cos n\pi=(-1)^n$, $\sin n\pi=0$, and by (c) the sum is $\pi^2/2$:

$$\frac{\pi^2}{2}=\frac{\pi^2}{6}+\sum_{n=1}^{\infty}\frac{2(-1)^n(-1)^n}{n^2}=\frac{\pi^2}{6}+2\sum_{n=1}^{\infty}\frac{1}{n^2}\implies\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{1}{2}\Big(\frac{\pi^2}{2}-\frac{\pi^2}{6}\Big)=\frac{\pi^2}{6}.$$

Step 3. Consistency check

Subtracting the two series, the odd terms cancel and the even ones double: $\sum \frac{1}{n^2} - \sum \frac{(-1)^{n+1}}{n^2} = 2\sum_{m\geq 1} \frac{1}{(2m)^2} = \frac{1}{2}\sum_{m\geq 1} \frac{1}{m^2}$. With the values found: $\frac{\pi^2}{6} - \frac{\pi^2}{12} = \frac{\pi^2}{12}$ and $\frac{1}{2}\cdot\frac{\pi^2}{6} = \frac{\pi^2}{12}$. They agree.

KEY RESULTS

- $x = 0$: $0 = \dfrac{\pi^2}{6} + 2\sum \dfrac{(-1)^n}{n^2}$, so $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$.
- $x = \pi$: $\dfrac{\pi^2}{2} = \dfrac{\pi^2}{6} + 2\sum \dfrac{1}{n^2}$, so $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.
- Consistency: $\dfrac{\pi^2}{6} - \dfrac{\pi^2}{12} = 2\displaystyle\sum_{n \text{ even}} \frac{1}{n^2} = \frac{1}{2}\cdot\frac{\pi^2}{6}$.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $a_n = O(1/n^2)$ because f is continuous at 0, while $b_n \sim (-1)^{n+1}\pi/n$ decays only like $1/n$: that slow part is produced by the jump of size π^2 at $x = \pm\pi$ (a jump of size J at $x = \pi$ always gives a contribution $\frac{J}{\pi}\frac{(-1)^n}{n}$ with $J = -\pi^2$ here). Near the jump the partial sums overshoot by about 9% of the jump (Gibbs phenomenon, **extension**: the overshoot of partial sums near a jump, which does not vanish as $N \to \infty$).
- **Checks:** the antiderivatives of Steps 3 and 4 give back $x^2\cos nx$ and $x^2\sin nx$ when differentiated; partial sums with 4000 terms reproduce 0 at $x = 0$, $\pi^2/2$ at $x = \pi$ and $\pi^2/4$ at $x = \pi/2$ to 10^{-3}; the mean value $a_0/2 = \pi^2/6$ equals $\frac{1}{2\pi}\int_{-\pi}^{\pi} f\,dx$.
- **Common mistakes in the exam:** (1) dividing by 2π instead of π in a_n, or forgetting that the constant term is $a_0/2$; (2) evaluating the series at $x = \pi$ as $f(\pi^-) = \pi^2$ instead of the midpoint $\pi^2/2$, which gives $\sum 1/n^2 = 5\pi^2/12$ (wrong); (3) dropping the term $2((-1)^n - 1)/(\pi n^3)$ of b_n, which only lives on odd n; (4) treating f as even or odd and discarding one family of coefficients.

2nd Midterm · Unit 5 · Fourier series and convergence · 5 points · 45 min · Standard exam level

P11 — Half-range sine and cosine series of f(x) = x^2 on (0, pi): continuity of the extensions, decay of the coefficients, values of the series and three numerical sums

Let $f(x) = x^2$ for $0 < x < \pi$.

*(a) Compute the Fourier **sine** series of f on $(0, \pi)$, writing every integration by parts. Give b_n separately for odd and even n.*

*(b) Compute the Fourier **cosine** series of f on $(0, \pi)$.*

*(c) Which of the two periodic extensions (odd or even, of period 2π) is continuous on $\mathbb{R}$? Explain why this decides how fast the coefficients of (a) and (b) decay. Then give, with justification, the value of **each** series at $x = \pi$, $x = -\pi/2$ and $x = 5\pi/2$.*

(d) Use the series to obtain

$$\sum_{n=1}^{\infty} \frac{1}{n^2}, \qquad \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}, \qquad \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^3} = 1 - \frac{1}{27} + \frac{1}{125} - \cdots$$

You may use the Leibniz sum $\displaystyle\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \frac{\pi}{4}$.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Data: $f(x) = x^2$ on the half-period $(0, L)$ with $L = \pi$. f is smooth on $[0, \pi]$ with $f(0) = 0$, $f(\pi) = \pi^2$, $f'(x) = 2x$, so $f'(0) = 0$, $f'(\pi) = 2\pi$.
- Unknowns: the coefficients b_n (sine series) and a_0, a_n (cosine series), and the pointwise sums of both series on all of $\mathbb{R}$.
- Regularity: f and f' are continuous on $[0, \pi]$, so both extensions are piecewise smooth and the convergence theorem (Haberman Sec. 3.2) applies: each series converges to the extension where it is continuous and to the average of the side limits at a jump.

Phase 2. Canonical formulation

With $L = \pi$ (Haberman Sec. 3.3):

$$f(x) \sim \sum_{n=1}^{\infty} b_n \sin nx, \qquad b_n = \frac{2}{\pi} \int_0^{\pi} x^2 \sin nx \, dx$$

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx, \qquad a_n = \frac{2}{\pi}\int_0^{\pi} x^2 \cos nx\, dx \quad (n \geq 0)$$

The sine series represents the **odd** 2π-periodic extension f_o of f; the cosine series represents the **even** 2π-periodic extension f_e.

Part (a) · 1.5 points

Compute the Fourier sine series of f on (0, pi)

Step 1. First integration by parts

Take $u = x^2$, $dv = \sin nx\, dx$, so $du = 2x\, dx$, $v = -\dfrac{\cos nx}{n}$:

$$\int x^2 \sin nx\, dx = -\frac{x^2 \cos nx}{n} + \frac{2}{n}\int x \cos nx\, dx$$

Step 2. Second integration by parts

Take $u = x$, $dv = \cos nx\, dx$, so $du = dx$, $v = \dfrac{\sin nx}{n}$:

$$\int x \cos nx\, dx = \frac{x \sin nx}{n} - \frac{1}{n}\int \sin nx\, dx = \frac{x \sin nx}{n} + \frac{\cos nx}{n^2}$$

Hence the antiderivative

$$F(x) = -\frac{x^2 \cos nx}{n} + \frac{2x \sin nx}{n^2} + \frac{2 \cos nx}{n^3}$$

Check: $F'(x) = -\dfrac{2x \cos nx}{n} + x^2 \sin nx + \dfrac{2 \sin nx}{n^2} + \dfrac{2x \cos nx}{n} - \dfrac{2 \sin nx}{n^2} = x^2 \sin nx.$

Step 3. Barrow's rule

$F(\pi) = -\dfrac{\pi^2(-1)^n}{n} + 0 + \dfrac{2(-1)^n}{n^3}$ and $F(0) = 0 + 0 + \dfrac{2}{n^3}$, so

$$\int_0^{\pi} x^2 \sin nx \, dx = [F]_0^{\pi} = F(\pi) - F(0) = \frac{\pi^2(-1)^{n+1}}{n} + \frac{2\big((-1)^n - 1\big)}{n^3}$$

Multiplying by $2/\pi$:

$$b_n = \frac{2\pi(-1)^{n+1}}{n} + \frac{4\big((-1)^n - 1\big)}{\pi n^3}$$

Step 4. Odd and even indices

For odd n, $(-1)^{n+1} = 1$ and $(-1)^n - 1 = -2$: $b_n = \dfrac{2\pi}{n} - \dfrac{8}{\pi n^3}$. For even n, $(-1)^{n+1} = -1$ and $(-1)^n - 1 = 0$: $b_n = -\dfrac{2\pi}{n}$. The leading term is of order $1/n$ for every n.

KEY RESULTS

- $b_n = \dfrac{2\pi(-1)^{n+1}}{n} + \dfrac{4\big((-1)^n - 1\big)}{\pi n^3}$.
- Odd n: $b_n = \dfrac{2\pi}{n} - \dfrac{8}{\pi n^3}$; even n: $b_n = -\dfrac{2\pi}{n}$. For example $b_1 = 2\pi - \dfrac{8}{\pi}$, $b_2 = -\pi$.
- $x^2 = \displaystyle\sum_{n=1}^{\infty} \left[\frac{2\pi(-1)^{n+1}}{n} + \frac{4((-1)^n - 1)}{\pi n^3}\right] \sin nx$ for $0 < x < \pi$.

Part (b) · 1 point

Compute the Fourier cosine series of f on (0, pi)

Step 1. Mean value

$$a_0 = \frac{2}{\pi}\int_0^{\pi} x^2 \, dx = \frac{2}{\pi}\left[\frac{x^3}{3}\right]_0^{\pi} = \frac{2}{\pi}\cdot\frac{\pi^3}{3} = \frac{2\pi^2}{3}$$

Step 2. Antiderivative

With $u = x^2$, $dv = \cos nx\,dx$ ($du = 2x\,dx$, $v = \sin nx/n$), then $u = x$, $dv = \sin nx\,dx$ ($du = dx$, $v = -\cos nx/n$):

$$\int x^2 \cos nx\,dx = \frac{x^2 \sin nx}{n} - \frac{2}{n}\int x \sin nx\,dx = \frac{x^2 \sin nx}{n} - \frac{2}{n}\left(-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\right)$$

$$G(x) = \frac{x^2 \sin nx}{n} + \frac{2x\cos nx}{n^2} - \frac{2\sin nx}{n^3}$$

Step 3. Barrow's rule

$G(\pi) = 0 + \dfrac{2\pi(-1)^n}{n^2} - 0$, $G(0) = 0$. Therefore

$$a_n = \frac{2}{\pi}\,[G]_0^{\pi} = \frac{2}{\pi}\cdot\frac{2\pi(-1)^n}{n^2} = \frac{4(-1)^n}{n^2}$$

KEY RESULTS

- $a_0 = \dfrac{2\pi^2}{3}$, so the constant term is $\dfrac{a_0}{2} = \dfrac{\pi^2}{3}$.
- $a_n = \dfrac{4(-1)^n}{n^2}$ for $n \geq 1$.
- $x^2 = \dfrac{\pi^2}{3} + 4\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}\cos nx$ for $0 \leq x \leq \pi$.

Part (c) · 1.25 points

Continuity of the extensions, decay of the coefficients and values of the series

Step 1. The odd extension

$f_o(x) = x^2$ on $(0, \pi)$, $f_o(x) = -x^2$ on $(-\pi, 0)$, repeated with period 2π. At $x = 0$ both side limits are 0: continuous. At $x = \pi$: $f_o(\pi^-) = \pi^2$, while $f_o(\pi^+) = f_o(-\pi^+) = -(\pi)^2 = -\pi^2$. Since $f(\pi) \neq 0$, the odd extension has a jump of size $2\pi^2$ at every odd multiple of π.

Step 2. The even extension

$f_e(x) = x^2$ on $[-\pi, \pi]$, repeated with period 2π. At $x = \pi$: $f_e(\pi^-) = \pi^2 = f_e(-\pi^+) = f_e(\pi^+)$: continuous. The even extension of a continuous function is always continuous. Its derivative is $2x$ on $(-\pi, \pi)$: $f_e'(\pi^-) = 2\pi$ and $f_e'(\pi^+) = f_e'(-\pi^+) = -2\pi$, a jump; at $x = 0$, $f_e'(0^\pm) = 0$, no jump.

Step 3. Why continuity controls the decay

In (a), the term of order $1/n$ came from the boundary value $-\dfrac{x^2 \cos nx}{n}\Big|_0^\pi$, which is nonzero exactly because $f(\pi) \neq 0$, i.e. because f_o jumps. In (b) the corresponding boundary term $\dfrac{x^2 \sin nx}{n}\Big|_0^\pi$ vanishes for any f; the first surviving term is $\left[\dfrac{2x \cos nx}{n^2}\right]_0^\pi = \dfrac{f'(\pi)(-1)^n - f'(0)}{n^2}$, which is nonzero because f_e' jumps at $x = \pi$ ($f'(\pi) = 2\pi \neq 0$). Thus a jump in the extension forces $1/n$; a continuous extension with a jump in its derivative gives $1/n^2$. This matches Haberman Sec. 3.4: the cosine series of a continuous f can be differentiated term by term (giving the sine series of $f' = 2x$), while the sine series of x^2 cannot, because f_o is not continuous.

Step 4. Values of the sine series

By the convergence theorem the sine series converges to $\frac{1}{2}\left[f_o(x^+) + f_o(x^-)\right]$:

- $x = \pi$: $\frac{1}{2}(\pi^2 - \pi^2) = 0$. Indeed every term $b_n \sin n\pi = 0$.
- $x = -\pi/2$: f_o continuous there, $f_o(-\pi/2) = -(\pi/2)^2 = -\pi^2/4$.
- $x = 5\pi/2 = \pi/2 + 2\pi$: by periodicity $f_o(\pi/2) = \pi^2/4$.

Step 5. Values of the cosine series

f_e is continuous, so the cosine series equals f_e everywhere:

- $x = \pi$: π^2.
- $x = -\pi/2$: $f_e(-\pi/2) = (\pi/2)^2 = \pi^2/4$.
- $x = 5\pi/2$: $f_e(\pi/2) = \pi^2/4$.

KEY RESULTS

- Odd extension f_o: **discontinuous** at $x = \pm\pi, \pm 3\pi, \ldots$ (jump from π^2 to $-\pi^2$), so $b_n = O(1/n)$.
- Even extension f_e: **continuous** everywhere; only f_e' jumps (from 2π to -2π at odd multiples of π), so $a_n = O(1/n^2)$.
- Sine series: 0 at $x = \pi$, $-\pi^2/4$ at $x = -\pi/2$, $\pi^2/4$ at $x = 5\pi/2$.
- Cosine series: π^2 at $x = \pi$, $\pi^2/4$ at $x = -\pi/2$, $\pi^2/4$ at $x = 5\pi/2$.

Part (d) · 1.25 points

Numerical sums from the series

Step 1. Cosine series at x = pi

$\cos n\pi = (-1)^n$, so $(-1)^n \cos n\pi = 1$ and

$$\pi^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{1}{n^2} \;\Rightarrow\; \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{4}\left(\pi^2 - \frac{\pi^2}{3}\right) = \frac{\pi^2}{6}$$

Step 2. Cosine series at x = 0

$\cos 0 = 1$ and $f_e(0) = 0$:

$$0 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \;\Rightarrow\; \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$$

Step 3. Sine series at x = pi/2

$x = \pi/2$ is a point of continuity of f_o, so the series equals $\pi^2/4$. Only odd $n = 2k+1$ survive, with $\sin\big((2k+1)\pi/2\big) = (-1)^k$ and $b_{2k+1} = \dfrac{2\pi}{2k+1} - \dfrac{8}{\pi(2k+1)^3}$:

$$\frac{\pi^2}{4} = 2\pi \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} - \frac{8}{\pi}\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^3} = 2\pi \cdot \frac{\pi}{4} - \frac{8}{\pi} S_3$$

where S_3 is the requested sum. Splitting the series is allowed because both pieces converge. Then

$$\frac{8}{\pi} S_3 = \frac{\pi^2}{2} - \frac{\pi^2}{4} = \frac{\pi^2}{4} \;\Rightarrow\; S_3 = \frac{\pi^3}{32} \approx 0.968946$$

KEY RESULTS

- $\sum_{n\geq 1} \dfrac{1}{n^2} = \dfrac{\pi^2}{6}$ (cosine series at $x = \pi$).
- $\sum_{n\geq 1} \dfrac{(-1)^{n+1}}{n^2} = \dfrac{\pi^2}{12}$ (cosine series at $x = 0$).
- $\sum_{k\geq 0} \dfrac{(-1)^k}{(2k+1)^3} = \dfrac{\pi^3}{32} \approx 0.968946$ (sine series at $x = \pi/2$).

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** the sine series converges to 0 at $x = \pi$ although $f(\pi) = \pi^2$; near $x = \pi$ the partial sums overshoot (Gibbs phenomenon, extension: the overshoot of the partial sums near a jump does not vanish as more terms are added, it only moves closer to the jump). The cosine series converges uniformly, since $\sum 4/n^2 < \infty$.
- **Checks:** $S_3 = 1 - 0.037037 + 0.008 + \ldots$: the first three terms give 0.970963, already within 0.0021 of $\pi^3/32$. Differentiating the cosine series term by term gives $2x = \sum \frac{4(-1)^{n+1}}{n} \sin nx$, the sine series of $2x$ (as Haberman Sec. 3.4 allows for a continuous even extension).
- **Common mistakes in the exam:** (1) using the general a_n formula with $n = 0$ (division by zero) instead of computing a_0 apart; (2) forgetting the factor $\frac{1}{2}$ in $a_0/2$, which gives the wrong constant $2\pi^2/3$; (3) claiming the sine series equals π^2 at $x = \pi$; (4) evaluating at $x = -\pi/2$ with f instead of the extension (the sine series gives $-\pi^2/4$, not $+\pi^2/4$).

P12 — Heat equation with a constant source and fixed, different end temperatures: steady state, separation of variables for the deviation, series solution and the time constant of the slowest mode

A thin laterally insulated rod occupies $0 \le x \le L$*. Its ends are held at the constant temperatures* $u(0,t) = T_A$ *and* $u(L,t) = T_B$*. For* $t < 0$ *the rod has been left long enough to reach the equilibrium with no internal heat source. At* $t = 0$ *an electric current is switched on that heats the rod uniformly, so that for* $t > 0$ *the temperature* $u(x,t)$ *satisfies*

$$u_t = k\,u_{xx} + Q, \qquad 0 < x < L,\ t > 0,$$

with constant diffusivity $k > 0$ *and constant source* $Q > 0$ *(in K/s).*

(a) Write the initial temperature $u(x,0)$*. Find the steady state* $u_s(x)$ *of the problem for* $t > 0$ *and the steady temperature at the midpoint.*

(b) Write the problem satisfied by $v = u - u_s$*. Separate variables in it,* $v = X(x)T(t)$*, and discuss the three cases of the sign of the separation constant to find all eigenvalues and eigenfunctions.*

(c) Obtain $u(x,t)$ *as a series with all coefficients in closed form.*

(d) Find $\lim_{t\to\infty} u$*, the time constant* τ_1 *of the slowest mode, and, keeping only that mode, the time* t_{99} *after which the midpoint has completed* 99% *of its final rise. Numbers:* $L = 0.5$ *m,* $k = 1.0 \times 10^{-4}$ m^2 */s,* $Q = 0.02$ *K/s,* $T_A = 20$ *°C,* $T_B = 40$ *°C.*

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $u(x,t)$, independent variables $x \in [0,L]$, $t \ge 0$. One-dimensional conduction, constant properties, uniform volumetric source.
- Operator: linear, second order, parabolic; **non-homogeneous** twice: the source Q in the equation and the Dirichlet data T_A, T_B (non-zero). Separation of variables needs homogeneous boundary conditions, so the inhomogeneities are removed first with the steady state.
- Data: L [m], k [m^2/s], Q [K/s], T_A, T_B [°C or K].

Phase 2. Canonical formulation

$$u_t = ku_{xx} + Q, \quad u(0,t) = T_A, \quad u(L,t) = T_B, \quad u(x,0) = u_0(x)$$

Split $u(x,t) = u_s(x) + v(x,t)$, where u_s absorbs the source and the boundary data and v solves a homogeneous problem with homogeneous Dirichlet conditions (Haberman Sec. 7.2).

Part (a) · 1 point

Initial temperature and steady state

Step 1. Initial profile

Before the source is switched on the rod is in equilibrium, $u_t = 0$, so $k u'' = 0$ and $u = \alpha + \beta x$. The end values give $\alpha = T_A$, $\alpha + \beta L = T_B$, hence $u_0(x) = T_A + (T_B - T_A)x/L$ (Haberman Sec. 1.4).

Step 2. Steady state with the source

$u_s'' = -\dfrac{Q}{k}$. Integrating once: $u_s' = -\dfrac{Q}{k}x + C_1$; again: $u_s = -\dfrac{Q}{2k}x^2 + C_1 x + C_2$.

Step 3. Constants

$u_s(0) = C_2 = T_A$. $u_s(L) = -\dfrac{QL^2}{2k} + C_1 L + T_A = T_B$, so $C_1 = \dfrac{T_B - T_A}{L} + \dfrac{QL}{2k}$. Substituting and grouping $\dfrac{Q}{2k}(Lx - x^2)$:

$$u_s(x) = T_A + (T_B - T_A)\frac{x}{L} + \frac{Q}{2k}\,x(L - x)$$

Check: $k u_s'' = k \cdot \dfrac{Q}{2k} \cdot (-2) = -Q$, and $x(L-x)$ vanishes at both ends.

Step 4. Midpoint

$u_s(L/2) = T_A + \dfrac{T_B - T_A}{2} + \dfrac{Q}{2k} \cdot \dfrac{L}{2} \cdot \dfrac{L}{2} = \dfrac{T_A + T_B}{2} + \dfrac{QL^2}{8k}$. Numbers: $\dfrac{0.02 \cdot 0.25}{8 \cdot 10^{-4}} = 6.25$ K, so $u_s(L/2) = 36.25$ °C.

KEY RESULTS

- $u_0(x) = T_A + (T_B - T_A)\dfrac{x}{L}$.
- $u_s(x) = T_A + (T_B - T_A)\dfrac{x}{L} + \dfrac{Q}{2k}\,x(L-x)$.
- $u_s(L/2) = \dfrac{T_A + T_B}{2} + \dfrac{QL^2}{8k}$; with the numbers $30 + 6.25 = 36.25$ °C.

Part (b) · 1.5 points

Problem for v and separation of variables

Step 1. Homogeneous problem

$v_t = u_t - 0 = ku_{xx} + Q - (ku_s'' + Q) = k(u - u_s)_{xx} = kv_{xx}$. At the ends $v(0,t) = T_A - T_A = 0$ and $v(L,t) = T_B - T_B = 0$. Initially $v(x,0) = u_0 - u_s = -\dfrac{Q}{2k}x(L-x)$: the linear parts cancel.

Step 2. Separation

$v = X(x)T(t)$ gives $XT' = kX''T$, and dividing by kXT:

$$\frac{T'}{kT} = \frac{X''}{X} = -\lambda$$

The left side depends only on t and the middle only on x, so both equal a constant. Hence $T' = -\lambda kT$ and $X'' + \lambda X = 0, X(0) = X(L) = 0$ (the conditions on v with $T \not\equiv 0$).

Step 3. Case lambda < 0

Write $\lambda = -\mu^2, \mu > 0$: $X = a\cosh \mu x + b\sinh \mu x$. $X(0) = a = 0$; $X(L) = b\sinh \mu L = 0$ with $\sinh \mu L > 0$, so $b = 0$. Only the trivial solution.

Step 4. Case lambda = 0

$X'' = 0, X = a + bx$. $X(0) = a = 0, X(L) = bL = 0 \Rightarrow b = 0$. Trivial.

Step 5. Case lambda > 0

$\lambda = \mu^2$: $X = a\cos \mu x + b\sin \mu x$. $X(0) = a = 0$; $X(L) = b\sin \mu L = 0$ with $b \neq 0$ requires $\mu L = n\pi$. Hence

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \qquad X_n(x) = \sin \frac{n\pi x}{L}, \qquad n = 1, 2, 3, \ldots$$

and $T' = -k\lambda_n T$ integrates to $T_n(t) = e^{-k\lambda_n t}$.

KEY RESULTS

- $v_t = kv_{xx}, v(0,t) = v(L,t) = 0, v(x,0) = u_0 - u_s = -\dfrac{Q}{2k}x(L-x)$.
- $\lambda < 0$ and $\lambda = 0$: only $X \equiv 0$.
- $\lambda_n = \left(\dfrac{n\pi}{L}\right)^2, X_n = \sin \dfrac{n\pi x}{L}, T_n = e^{-kn^2\pi^2 t/L^2}, n = 1, 2, \ldots$

Part (c) · 1.5 points

Series solution

Step 1. Superposition

Each product X_nT_n satisfies $v_t = kv_{xx}$ and the zero end conditions, and so does any convergent combination:

$$v(x,t) = \sum_{n=1}^{\infty} c_n \sin\frac{n\pi x}{L}\, e^{-kn^2\pi^2 t/L^2}$$

At $t=0$: $-\dfrac{Q}{2k}x(L-x) = \sum c_n \sin\dfrac{n\pi x}{L}$, so by orthogonality $c_n = \dfrac{2}{L}\displaystyle\int_0^L \Big(-\frac{Q}{2k}\Big)x(L-x)\sin\frac{n\pi x}{L}\,dx$.

Step 2. The integral

Write $\omega = n\pi/L$. With $u = x(L-x) = Lx - x^2$, $dv = \sin\omega x\,dx$ ($du = (L-2x)\,dx$, $v = -\cos\omega x/\omega$):

$$\int_0^L (Lx-x^2)\sin\omega x\,dx = \Big[-\frac{(Lx-x^2)\cos\omega x}{\omega}\Big]_0^L + \frac{1}{\omega}\int_0^L (L-2x)\cos\omega x\,dx$$

The bracket is 0 because $Lx-x^2$ vanishes at $x=0$ and $x=L$. Next, $u = L-2x$, $dv = \cos\omega x\,dx$ ($du = -2\,dx$, $v = \sin\omega x/\omega$):

$$\int_0^L (L-2x)\cos\omega x\,dx = \Big[\frac{(L-2x)\sin\omega x}{\omega}\Big]_0^L + \frac{2}{\omega}\int_0^L \sin\omega x\,dx = 0 + \frac{2}{\omega}\Big[-\frac{\cos\omega x}{\omega}\Big]_0^L = \frac{2}{\omega^2}\big(1-\cos n\pi\big)$$

since $\sin\omega L = \sin n\pi = 0$. Hence $\displaystyle\int_0^L x(L-x)\sin\omega x\,dx = \frac{2(1-(-1)^n)}{\omega^3} = \frac{2L^3(1-(-1)^n)}{n^3\pi^3}$ and

$$\frac{2}{L}\int_0^L x(L-x)\sin\frac{n\pi x}{L}\,dx = \frac{4L^2(1-(-1)^n)}{n^3\pi^3}$$

Step 3. Coefficients and solution

Multiplying by $-\dfrac{Q}{2k}$: $c_n = -\dfrac{2QL^2(1-(-1)^n)}{k\pi^3 n^3}$, which is $-\dfrac{4QL^2}{k\pi^3 n^3}$ for odd n and 0 for even n. Adding u_s :

$$u(x,t) = T_A + (T_B - T_A)\frac{x}{L} + \frac{Q}{2k}x(L-x) - \frac{4QL^2}{k\pi^3}\sum_{n=1,3,5,\dots}\frac{1}{n^3}\sin\frac{n\pi x}{L}\,e^{-kn^2\pi^2 t/L^2}$$

The coefficients decay as $1/n^3$ (the odd extension of $x(L-x)$ is continuous with a continuous derivative), so the series converges fast even at $t = 0$.

KEY RESULTS

- $\dfrac{2}{L}\displaystyle\int_0^L x(L-x)\sin\frac{n\pi x}{L}\,dx = \frac{4L^2}{n^3\pi^3}\bigl(1-(-1)^n\bigr)$: $\dfrac{8L^2}{n^3\pi^3}$ for odd n, 0 for even n.
- $c_n = -\dfrac{4QL^2}{k\pi^3 n^3}$ for odd n, $c_n = 0$ for even n.
- $u(x,t) = T_A + (T_B - T_A)\dfrac{x}{L} + \dfrac{Q}{2k}x(L-x) - \dfrac{4QL^2}{k\pi^3}\displaystyle\sum_{n\ \text{odd}}\frac{1}{n^3}\sin\frac{n\pi x}{L}\,e^{-kn^2\pi^2 t/L^2}$.

Part (d) · 1 point

Long-time limit, time constant and the 99 % time

Step 1. Limit

Every term carries $e^{-kn^2\pi^2 t/L^2} \to 0$ and $\sum 1/n^3$ converges, so $v \to 0$ and $u \to u_s$: the rod forgets its initial state.

Step 2. Slowest mode

The $n = 1$ term decays as e^{-t/τ_1} with $\tau_1 = \dfrac{1}{k\lambda_1} = \dfrac{L^2}{k\pi^2} = \dfrac{0.25}{10^{-4}\cdot 9.8696} = 253.3$ s. The next non-zero mode, $n = 3$, has $\tau_3 = \tau_1/9$.

Step 3. Midpoint rise

The total rise of the midpoint is $u_s(L/2) - u_0(L/2) = \dfrac{QL^2}{8k} = 6.25$ K. At $x = L/2$ the deficit is

$$u_s\Bigl(\frac{L}{2}\Bigr) - u\Bigl(\frac{L}{2},t\Bigr) = \frac{4QL^2}{k\pi^3}\sum_{m=0}^{\infty}\frac{(-1)^m}{(2m+1)^3}e^{-(2m+1)^2 t/\tau_1}$$

At $t=0$ the sum is $\pi^3/32$ (bank item P11), giving $\dfrac{4QL^2}{k\pi^3}\cdot\dfrac{\pi^3}{32}=\dfrac{QL^2}{8k}$, as it must. Keeping $m=0$, the fraction still missing is $\dfrac{32}{\pi^3}e^{-t/\tau_1}$.

Step 4. The 99 % time

$\dfrac{32}{\pi^3}e^{-t_{99}/\tau_1}=0.01 \Rightarrow t_{99}=\tau_1\ln\dfrac{3200}{\pi^3}=\tau_1\ln(103.2)=4.637\,\tau_1\approx 1174.5$ s. At that time the $m=1$ term is smaller than the $m=0$ term by the factor $\frac{1}{27}e^{-8\cdot 4.637}<10^{-17}$: the one-mode answer is exact for practical purposes.

KEY RESULTS

- $u(x,t)\to u_s(x)$ as $t\to\infty$.
- $\tau_1=\dfrac{L^2}{k\pi^2}=253.3$ s; the $n=3$ mode decays 9 times faster.
- Midpoint: $\dfrac{u_s(L/2)-u(L/2,t)}{QL^2/(8k)}\approx\dfrac{32}{\pi^3}e^{-t/\tau_1}$, so $t_{99}=\tau_1\ln\dfrac{3200}{\pi^3}=4.637\,\tau_1\approx 1174.5$ s (about 19.6 min).

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $Q\to 0$ gives $u_s=u_0$ and $v\equiv 0$: nothing happens, as expected. $t\to\infty$: $u\to u_s$, a parabola above the straight line between T_A and T_B; with these numbers its maximum is interior, at $x^*=\dfrac{L}{2}+\dfrac{k(T_B-T_A)}{QL}=0.25+0.2=0.45$ m, since $u_s'(x^*)=0$.
- **Checks:** dimensions: $\dfrac{Q}{k}x^2$ is $\dfrac{\mathrm{K\,s^{-1}}}{\mathrm{m^2 s^{-1}}}\mathrm{m^2}=\mathrm{K}$; $\tau_1=L^2/k$ is s. Each mode satisfies $v_t=kv_{xx}$ and vanishes at both ends; the series at $t=0$ reproduces u_0 (checked numerically in the script).
- **Common mistakes in the exam:** (1) separating variables directly in u, whose boundary conditions are not homogeneous; (2) taking $v(x,0)=u_0$ instead of u_0-u_s; (3) skipping the cases $\lambda\le 0$; (4) giving the time constant as L^2/k (forgetting π^2) or using the decay rate of v^2.

2nd Midterm · Unit 6 · Laplace equation by separation · 5 points · 45 min · Standard exam level

P13 — Laplace's equation in a rectangle with Neumann conditions on the vertical sides and Dirichlet data given by finite cosine sums: the zero mode, cosh and sinh modes, flux balance and uniqueness

The steady temperature $u(x,y)$ of the plate $0<x<\pi$, $0<y<1$ satisfies Laplace's equation

$$u_{xx}+u_{yy}=0.$$

The vertical sides are insulated and the horizontal sides are held at prescribed temperatures:

$$u_x(0,y)=u_x(\pi,y)=0, \qquad u(x,0)=2+\cos 2x, \qquad u(x,1)=4+3\cos x.$$

(a) Separate variables, $u=X(x)Y(y)$, and solve the eigenvalue problem for X, discussing the three signs of the separation constant.

(b) Solve the equations for Y, paying special attention to the zero eigenvalue, and write the general solution that satisfies the conditions on the insulated sides.

(c) Impose the conditions on $y=0$ and $y=1$ and obtain $u(x,y)$ in closed form. Give u at the centre $(\pi/2,\ 1/2)$.

(d) Compute $\Phi(y)=\displaystyle\int_0^{\pi} u_y(x,y)\,dx$ and explain why it does not depend on y. Is the solution unique? Would it be unique, and would it exist for any data, if the horizontal sides were also insulated?

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $u(x,y)$ on the rectangle $R=(0,\pi)\times(0,1)$; steady two-dimensional conduction, no sources.
- Operator: Laplacian, linear, second order, elliptic, homogeneous equation. The conditions are homogeneous (Neumann) in x and non-homogeneous (Dirichlet) in y, so the eigenvalue problem is posed in x.
- Data: the bottom and top temperatures are **finite cosine sums**, i.e. already combinations of the eigenfunctions found in (a); no integral is needed for the coefficients.

Phase 2. Canonical formulation

$$\Delta u = 0 \text{ in } R, \qquad \frac{\partial u}{\partial x} = 0 \text{ on } x = 0, \pi, \qquad u = f(x) \text{ on } y = 0, \qquad u = g(x) \text{ on } y = 1$$

with $f(x) = 2 + \cos 2x$ and $g(x) = 4 + 3\cos x$. Sturm-Liouville problem in x: $X'' + \lambda X = 0, X'(0) = X'(\pi) = 0$.

Part (a) · 1.5 points

Separation and the eigenvalue problem in x

Step 1. Separation

$u = X(x)Y(y)$ gives $X''Y + XY'' = 0$. Dividing by XY: $\dfrac{X''}{X} = -\dfrac{Y''}{Y}$. The left side depends only on x, the right only on y; both equal a constant $-\lambda$:

$$X'' + \lambda X = 0, \qquad Y'' - \lambda Y = 0$$

The side conditions $u_x(0, y) = X'(0)Y(y) = 0$ for all y and $Y \not\equiv 0$ give $X'(0) = 0$; likewise $X'(\pi) = 0$.

Step 2. Case lambda < 0

$\lambda = -m^2, m > 0$: $X = a\cosh mx + b\sinh mx$, $X' = m(a\sinh mx + b\cosh mx)$. $X'(0) = mb = 0 \Rightarrow b = 0$. $X'(\pi) = ma\sinh m\pi = 0$ with $\sinh m\pi > 0 \Rightarrow a = 0$. Trivial.

Step 3. Case lambda = 0

$X'' = 0, X = a + bx, X' = b$. Both conditions give $b = 0$, and a is free: $X_0 = 1$ is an eigenfunction. With Neumann conditions the zero eigenvalue **does** occur (Haberman Sec. 2.4).

Step 4. Case lambda > 0

$\lambda = m^2$: $X = a\cos mx + b\sin mx$, $X' = m(-a\sin mx + b\cos mx)$. $X'(0) = mb = 0 \Rightarrow b = 0$.
$X'(\pi) = -ma\sin m\pi = 0$ with $a \neq 0$ requires $m = n \in \{1, 2, \dots\}$. So $\lambda_n = n^2, X_n = \cos nx$.

KEY RESULTS

- $\dfrac{X''}{X} = -\dfrac{Y''}{Y} = -\lambda$; $X'' + \lambda X = 0, X'(0) = X'(\pi) = 0$; $Y'' - \lambda Y = 0$.
- $\lambda < 0$: only $X \equiv 0$.
- $\lambda_0 = 0$ is an eigenvalue with $X_0 = 1$.
- $\lambda_n = n^2, X_n = \cos nx, n = 1, 2, \dots$

Part (b) · 1 point

The y equations and the general solution

Step 1. Zero mode

$\lambda = 0$: $Y'' = 0$, integrate twice: $Y' = B_0$, $Y = A_0 + B_0 y$. The product $X_0 Y_0 = A_0 + B_0 y$ is the part of the temperature that varies linearly between the plates.

Step 2. Modes n >= 1

$Y'' - n^2 Y = 0$ has characteristic equation $r^2 - n^2 = 0$, $r = \pm n$, so $Y = \alpha e^{ny} + \beta e^{-ny}$. With $\cosh ny = \frac{1}{2}(e^{ny} + e^{-ny})$ and $\sinh ny = \frac{1}{2}(e^{ny} - e^{-ny})$ this is $Y_n = A_n \cosh ny + B_n \sinh ny$ ($A_n = \alpha + \beta$, $B_n = \alpha - \beta$). The cosh / sinh form is preferred because $\sinh 0 = 0$ and $\cosh 0 = 1$ make the condition at $y = 0$ immediate.

Step 3. Superposition

Every product satisfies Laplace's equation and the side conditions, and so does the sum:

$$u(x, y) = A_0 + B_0 y + \sum_{n=1}^{\infty} \cos nx \left(A_n \cosh ny + B_n \sinh ny \right)$$

KEY RESULTS

- $Y_0 = A_0 + B_0 y$ (linear in y).
- $Y_n = A_n \cosh ny + B_n \sinh ny$.
- $u(x, y) = A_0 + B_0 y + \sum_{n=1}^{\infty} \cos nx \left(A_n \cosh ny + B_n \sinh ny \right)$.

Part (c) · 1.5 points

Coefficients from the data and the closed form

Step 1. Bottom side

$u(x, 0) = A_0 + \sum_n A_n \cos nx = 2 + \cos 2x$. By orthogonality of $\{1, \cos nx\}$ on $(0, \pi)$ the coefficients are unique: $A_0 = 2$, $A_2 = 1$, $A_n = 0$ for $n \neq 2$. (Directly: $\frac{1}{\pi} \int_0^{\pi} (2 + \cos 2x) dx = 2$ and $\frac{2}{\pi} \int_0^{\pi} (2 + \cos 2x) \cos nx \, dx = \delta_{n2}$.)

Step 2. Top side

$u(x,1) = A_0 + B_0 + \sum_n (A_n \cosh n + B_n \sinh n)\cos nx = 4 + 3\cos x$:

- constant: $2 + B_0 = 4 \Rightarrow B_0 = 2$;
- $n = 1$: $0 \cdot \cosh 1 + B_1 \sinh 1 = 3 \Rightarrow B_1 = \dfrac{3}{\sinh 1}$;
- $n = 2$: $1 \cdot \cosh 2 + B_2 \sinh 2 = 0 \Rightarrow B_2 = -\dfrac{\cosh 2}{\sinh 2} = -\coth 2$;
- $n \geq 3$: $B_n \sinh n = 0 \Rightarrow B_n = 0$.

Step 3. Simplify the n = 2 mode

$$\cosh 2y - \frac{\cosh 2}{\sinh 2}\sinh 2y = \frac{\sinh 2\cosh 2y - \cosh 2 \sinh 2y}{\sinh 2} = \frac{\sinh(2-2y)}{\sinh 2}$$

by $\sinh(a-b) = \sinh a \cosh b - \cosh a \sinh b$. It equals 1 at $y = 0$ and 0 at $y = 1$, as required. Therefore

$$u(x,y) = 2 + 2y + \frac{3\sinh y}{\sinh 1}\cos x + \frac{\sinh\big(2(1-y)\big)}{\sinh 2}\cos 2x$$

Step 4. Check

$u_{xx} = -\dfrac{3\sinh y}{\sinh 1}\cos x - \dfrac{4\sinh(2-2y)}{\sinh 2}\cos 2x$ and $u_{yy} = \dfrac{3\sinh y}{\sinh 1}\cos x + \dfrac{(-2)^2\sinh(2-2y)}{\sinh 2}\cos 2x$ (chain rule: $\frac{d}{dy}\sinh(2-2y) = -2\cosh(2-2y)$), so $u_{xx} + u_{yy} = 0$. u_x contains $\sin x$ and $\sin 2x$, both zero at $x = 0, \pi$.

Step 5. Centre

$\cos(\pi/2) = 0$, $\cos\pi = -1$: $u(\pi/2, 1/2) = 2 + 1 - \dfrac{\sinh 1}{\sinh 2}$. With $\sinh 2 = 2\sinh 1 \cosh 1$:

$$u\Big(\frac{\pi}{2}, \frac{1}{2}\Big) = 3 - \frac{1}{2\cosh 1} \approx 3 - 0.324027 = 2.675973$$

KEY RESULTS

- $A_0 = 2, B_0 = 2$; $A_1 = 0, B_1 = \dfrac{3}{\sinh 1}$; $A_2 = 1, B_2 = -\coth 2$; all other coefficients 0.
- $u(x,y) = 2 + 2y + \dfrac{3\sinh y}{\sinh 1}\cos x + \dfrac{\sinh\big(2(1-y)\big)}{\sinh 2}\cos 2x$.
- $u(\pi/2, 1/2) = 3 - \dfrac{1}{2\cosh 1} \approx 2.675973$.

Part (d) · 1 point

Flux balance and uniqueness

Step 1. The integral

$u_y = 2 + \dfrac{3\cosh y}{\sinh 1}\cos x - \dfrac{2\cosh(2-2y)}{\sinh 2}\cos 2x$. Since $\int_0^\pi \cos x\,dx = [\sin x]_0^\pi = 0$ and $\int_0^\pi \cos 2x\,dx = \left[\tfrac{1}{2}\sin 2x\right]_0^\pi = 0$:

$$\Phi(y) = \int_0^\pi 2\,dx = 2\pi$$

Step 2. Why it is constant

$\Phi'(y) = \displaystyle\int_0^\pi u_{yy}\,dx = -\int_0^\pi u_{xx}\,dx = -\big[u_x\big]_{x=0}^{x=\pi} = -(0-0) = 0$. Physically the heat flux $-K\,u_y$ through any horizontal line is the same: there are no sources inside and the sides are insulated. The temperature rises linearly on average, from mean 2 at the bottom to mean 4 at the top, so $B_0 = 2$ is exactly the mean gradient.

Step 3. Uniqueness (Dirichlet on two sides)

Let u_1, u_2 be two solutions and $w = u_1 - u_2$: $\Delta w = 0$, $w_x = 0$ on $x = 0, \pi$, $w = 0$ on $y = 0, 1$. Green's first identity (extension: $\iint_R \left(w\Delta w + |\nabla w|^2\right)dA = \oint_{\partial R} w\,\dfrac{\partial w}{\partial n}\,ds$) gives $\iint_R |\nabla w|^2 dA = 0$, because on each side either $w = 0$ or $\partial w/\partial n = 0$. Hence $\nabla w = 0$, w is constant, and $w = 0$ on $y = 0$ makes it zero. The solution is unique. Consistently, the maximum principle (Haberman Sec. 2.5) bounds u between the extreme boundary values: $1 = u(\pi/2, 0) \le u \le u(0, 1) = 7$ (the maximum principle on this mixed problem is checked on a grid in the script).

Step 4. All sides insulated

If u_y were prescribed on $y = 0$ and $y = 1$ as well, the zero mode would only fix B_0 (twice) and leave A_0 free: solutions would differ by constants. Moreover, by Step 2 the prescribed fluxes would have to satisfy $\int_0^\pi u_y(x,0)\,dx = \int_0^\pi u_y(x,1)\,dx$ (solvability condition, Haberman Sec. 2.5): no net heat may enter a steady insulated plate. Data violating it give no solution at all.

KEY RESULTS

- $\Phi(y) = 2\pi$ for every y: only the zero mode carries net flux.
- Constant because, integrating $u_{xx} + u_{yy} = 0$ over $(0, \pi)$, $\Phi'(y) = -[u_x]_0^\pi = 0$: what enters through one plate leaves through the other (insulated sides).
- Unique: with Dirichlet data on two sides, $w \equiv 0$. With all four sides insulated, solutions exist only if the net flux data are compatible, and are unique only up to an additive constant.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** the $\cos 2x$ mode imposed at the bottom decays upward like $\sinh(2(1-y))/\sinh 2 \approx e^{-2y}$; higher modes $\cos nx$ would decay like e^{-ny}, so fine detail of boundary data is felt only near the side where it is imposed. The zero mode does not decay: it is the only part of the data felt across the whole plate.
- **Checks:** $u(x,0) = 2 + 0 + 0 + \cos 2x$ and $u(x,1) = 4 + 3\cos x + 0$; $u_x(0,y) = u_x(\pi,y) = 0$; the centre value 2.676 lies between the boundary extremes 1 and 7.
- **Common mistakes in the exam:** (1) discarding $\lambda = 0$ as "trivial", which loses the linear part $2 + 2y$ and makes the constant data impossible to match; (2) using $\sin nx$ eigenfunctions (those belong to Dirichlet sides); (3) writing $Y_n = A_n e^{ny} + B_n e^{-ny}$ and then solving a 2×2 system with large numbers instead of using $\sinh(n(1-y))$; (4) claiming uniqueness only up to a constant, which is true only when every side is insulated.

P14 — Vibrating string with fixed ends started from the rest position with a parabolic initial velocity: modes, coefficients from the velocity, d'Alembert check at the midpoint, energy of each mode and periodicity

A string of length L, linear density ρ and tension T_0 is fixed at $x = 0$ and $x = L$. Its small transverse displacement $u(x,t)$ satisfies

$$u_{tt} = c^2 u_{xx}, \qquad 0 < x < L,\ t > 0, \qquad c^2 = \frac{T_0}{\rho}.$$

At $t = 0$ the string lies along its rest position and is struck by a wide hammer that gives it the velocity profile

$$u(x,0) = 0, \qquad u_t(x,0) = g(x) = \frac{4V}{L^2}\, x(L - x),$$

where $V > 0$ is the speed given to the midpoint.

(a) Separate variables, discuss the sign of the separation constant and find the normal modes compatible with the fixed ends and with $u(x,0) = 0$.

(b) Obtain $u(x,t)$ as a series, with the coefficients in closed form.

(c) For $0 \le t \le L/(2c)$ use d'Alembert's formula to obtain the displacement of the midpoint, $u(L/2, t)$, in closed form. Check it against the series at $t = L/(2c)$ and deduce $\sum_{n\ \text{odd}} n^{-4}$.

(d) The energy of the string is $E(t) = \displaystyle\int_0^L \left(\frac{\rho}{2} u_t^2 + \frac{T_0}{2} u_x^2 \right) dx$. Compute the energy carried by each mode, check that their sum is the energy given by the hammer, and give the fraction carried by the fundamental. Show that the motion is periodic in time and describe the string at $t = L/(2c)$ and $t = L/c$.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $u(x,t)$ on $0 \le x \le L, t \ge 0$; small transverse vibrations, uniform string, no damping, no gravity.
- Operator: linear, second order, hyperbolic, homogeneous, with homogeneous Dirichlet conditions $u(0,t) = u(L,t) = 0$.
- Data: L [m], c [m/s], V [m/s], ρ [kg/m], $T_0 = \rho c^2$ [N]. Two initial conditions are needed (second order in t): zero displacement, non-zero velocity.

Phase 2. Canonical formulation

$$u_{tt} = c^2 u_{xx}, \quad u(0,t) = u(L,t) = 0, \quad u(x,0) = 0, \quad u_t(x,0) = g(x)$$

Eigenvalue problem in x: $X'' + \lambda X = 0$, $X(0) = X(L) = 0$. In t: $T'' + \lambda c^2 T = 0$, with $T(0) = 0$ from the zero initial displacement.

Part (a) · 1.25 points

Separation and normal modes

Step 1. Separation

$XT'' = c^2 X'' T$, so $\dfrac{T''}{c^2 T} = \dfrac{X''}{X} = -\lambda$ (each side depends on one variable only). Fixed ends: $X(0)T(t) = 0$ for all t, so $X(0) = 0$, and $X(L) = 0$.

Step 2. Case lambda < 0

$\lambda = -\mu^2$: $X = a\cosh \mu x + b \sinh \mu x$; $X(0) = a = 0$, $X(L) = b \sinh \mu L = 0 \Rightarrow b = 0$.

Step 3. Case lambda = 0

$X = a + bx$; $X(0) = a = 0$, $X(L) = bL = 0 \Rightarrow b = 0$.

Step 4. Case lambda > 0

$\lambda = \mu^2$: $X = a\cos \mu x + b \sin \mu x$; $X(0) = a = 0$; $b \sin \mu L = 0$ with $b \neq 0$ gives $\mu = n\pi/L$. Then $T'' + \omega_n^2 T = 0$ with $\omega_n = c\sqrt{\lambda_n} = \dfrac{n\pi c}{L}$, so $T = \alpha \cos \omega_n t + \beta \sin \omega_n t$. The condition $u(x,0) = X(x)T(0) = 0$ gives $\alpha = 0$. The normal modes are

$$u_n(x,t) = \sin \frac{n\pi x}{L} \, \sin \frac{n\pi c t}{L}, \qquad n = 1, 2, \ldots$$

KEY RESULTS

- $\dfrac{T''}{c^2 T} = \dfrac{X''}{X} = -\lambda$.
- $\lambda \le 0$: only $X \equiv 0$. $\lambda_n = \left(\dfrac{n\pi}{L}\right)^2$, $X_n = \sin\dfrac{n\pi x}{L}$.
- With $u(x,0) = 0$: $T_n = \sin \omega_n t$, $\omega_n = \dfrac{n\pi c}{L}$.

Part (b) · 1.5 points

Series solution with coefficients from the velocity

Step 1. Superposition and the velocity condition

$u = \sum_{n\ge 1} B_n \sin\dfrac{n\pi x}{L} \sin \omega_n t$ satisfies the equation, the fixed ends and $u(x,0) = 0$. Differentiating in t (chain rule: $\frac{d}{dt}\sin\omega_n t = \omega_n \cos\omega_n t$):

$$u_t(x,0) = \sum_{n\ge 1} \omega_n B_n \sin\frac{n\pi x}{L} = g(x) \;\Rightarrow\; \omega_n B_n = \frac{2}{L}\int_0^L g(x)\sin\frac{n\pi x}{L}\,dx$$

Step 2. The integral

Let $k = n\pi/L$. First parts, $u = Lx - x^2$, $dv = \sin kx\,dx$, $du = (L-2x)dx$, $v = -\cos kx/k$:

$$\int_0^L (Lx-x^2)\sin kx\,dx = \Big[-\frac{(Lx-x^2)\cos kx}{k}\Big]_0^L + \frac{1}{k}\int_0^L (L-2x)\cos kx\,dx = \frac{1}{k}\int_0^L (L-2x)\cos kx\,dx$$

Second parts, $u = L-2x$, $dv = \cos kx\,dx$, $du = -2dx$, $v = \sin kx/k$:

$$\int_0^L (L-2x)\cos kx\,dx = \Big[\frac{(L-2x)\sin kx}{k}\Big]_0^L + \frac{2}{k}\int_0^L \sin kx\,dx = 0 + \frac{2}{k}\Big[-\frac{\cos kx}{k}\Big]_0^L = \frac{2(1-(-1)^n)}{k^2}$$

So $\displaystyle\int_0^L x(L-x)\sin kx\,dx = \frac{2(1-(-1)^n)}{k^3} = \frac{2L^3(1-(-1)^n)}{\pi^3 n^3}$.

Step 3. Coefficients

$$\omega_n B_n = \frac{2}{L}\cdot\frac{4V}{L^2}\cdot\frac{2L^3(1-(-1)^n)}{\pi^3 n^3} = \frac{16V(1-(-1)^n)}{\pi^3 n^3}$$

which is $\dfrac{32V}{\pi^3 n^3}$ for odd n and 0 for even n (the even modes have a node at the midpoint and are not excited by a velocity profile symmetric about it). Dividing by $\omega_n = n\pi c/L$:

$$B_n = \frac{32VL}{c\pi^4 n^4}\ (n \text{ odd}), \qquad u(x,t) = \frac{32VL}{c\pi^4}\sum_{n=1,3,5,\ldots}\frac{1}{n^4}\sin\frac{n\pi x}{L}\sin\frac{n\pi ct}{L}$$

KEY RESULTS

- $\dfrac{2}{L}\displaystyle\int_0^L g(x)\sin\frac{n\pi x}{L}\,dx = \frac{32V}{\pi^3 n^3}$ for odd n, 0 for even n.
- $\omega_n B_n = \dfrac{32V}{\pi^3 n^3}$, so $B_n = \dfrac{32VL}{c\pi^4 n^4}$ for odd n.
- $u(x,t) = \dfrac{32VL}{c\pi^4}\displaystyle\sum_{n\ \text{odd}}\frac{1}{n^4}\sin\frac{n\pi x}{L}\sin\frac{n\pi ct}{L}$.

Part (c) · 1.25 points

d'Alembert check at the midpoint

Step 1. d'Alembert's formula

The general solution of $u_{tt} = c^2 u_{xx}$ is $F(x-ct) + H(x+ct)$; with $u(x,0) = 0$ and $u_t(x,0) = G(x)$ it becomes $u(x,t) = \dfrac{1}{2c}\displaystyle\int_{x-ct}^{x+ct} G(s)\,ds$ (extension: the course solves the string by separation of variables; d'Alembert's form is used here only as an independent check).

Step 2. The integral at the midpoint

With $s = L/2 + \sigma$, $ds = d\sigma$, the limits $s = L/2 \mp ct$ become $\sigma = \mp ct$, and $s(L-s) = \left(\dfrac{L}{2}+\sigma\right)\left(\dfrac{L}{2}-\sigma\right) = \dfrac{L^2}{4} - \sigma^2$, so $g = V - \dfrac{4V\sigma^2}{L^2}$:

$$u\left(\frac{L}{2},t\right) = \frac{1}{2c}\int_{-ct}^{ct}\left(V - \frac{4V\sigma^2}{L^2}\right)d\sigma = \frac{1}{2c}\left[V\sigma - \frac{4V\sigma^3}{3L^2}\right]_{-ct}^{ct} = \frac{1}{2c}\left(2Vct - \frac{8Vc^3t^3}{3L^2}\right)$$

$$u\Big(\frac{L}{2},t\Big)=Vt-\frac{4Vc^2t^3}{3L^2},\qquad 0\le t\le\frac{L}{2c}$$

For small t, $u\approx Vt$: the midpoint starts with speed V, as imposed.

Step 3. Comparison with the series

At $t=L/(2c)$: d'Alembert gives $\frac{VL}{2c}-\frac{4Vc^2}{3L^2}\cdot\frac{L^3}{8c^3}=\frac{VL}{2c}-\frac{VL}{6c}=\frac{VL}{3c}$. In the series, $\sin\frac{n\pi}{2}\sin\frac{n\pi}{2}=\big((-1)^{(n-1)/2}\big)^2=1$ for odd n:

$$\frac{VL}{3c}=\frac{32VL}{c\pi^4}\sum_{n\text{ odd}}\frac{1}{n^4}\;\Rightarrow\;\sum_{n\text{ odd}}\frac{1}{n^4}=\frac{\pi^4}{96}$$

KEY RESULTS

- $u(L/2,t)=Vt-\dfrac{4Vc^2t^3}{3L^2}$ for $0\le t\le\dfrac{L}{2c}$.
- At $t=L/(2c)$: $u=\dfrac{VL}{3c}$, and the series gives $\dfrac{32VL}{c\pi^4}\sum_{n\text{ odd}}n^{-4}$.
- $\displaystyle\sum_{n\text{ odd}}\frac{1}{n^4}=\frac{\pi^4}{96}$.

Part (d) · 1 point

Energy of the modes and periodicity

Step 1. Energy of one mode

For $u_n=B_n\sin k_nx\,\sin\omega_nt$ ($k_n=n\pi/L$): $\partial_tu_n=B_n\omega_n\sin k_nx\cos\omega_nt$, $\partial_xu_n=B_nk_n\cos k_nx\sin\omega_nt$. With $\int_0^L\sin^2k_nx\,dx=\int_0^L\cos^2k_nx\,dx=L/2$ and $T_0=\rho c^2$:

$$E_n=\frac{\rho}{2}B_n^2\omega_n^2\frac{L}{2}\cos^2\omega_nt+\frac{\rho c^2}{2}B_n^2k_n^2\frac{L}{2}\sin^2\omega_nt=\frac{\rho L}{4}\omega_n^2B_n^2$$

since $c^2k_n^2=\omega_n^2$. With $\omega_nB_n=32V/(\pi^3n^3)$: $E_n=\dfrac{\rho L}{4}\cdot\dfrac{1024V^2}{\pi^6n^6}=\dfrac{256\rho LV^2}{\pi^6n^6}$.

Step 2. Total energy

E is conserved (the ends do no work), so $E = E(0) = \dfrac{\rho}{2}\displaystyle\int_0^L \frac{16V^2}{L^4}x^2(L-x)^2\,dx$. Expanding $x^2(L-x)^2 = L^2x^2 - 2Lx^3 + x^4$ and integrating: $\left[\dfrac{L^2x^3}{3} - \dfrac{Lx^4}{2} + \dfrac{x^5}{5}\right]_0^L = L^5\left(\dfrac{1}{3} - \dfrac{1}{2} + \dfrac{1}{5}\right) = \dfrac{L^5}{30}$. Hence $E = \dfrac{8\rho V^2}{L^4}\cdot\dfrac{L^5}{30} = \dfrac{4\rho L V^2}{15}$. The modes are orthogonal, so $E = \sum E_n$, and with $\sum_{n\ \text{odd}} n^{-6} = \pi^6/960$ the sum indeed gives $\dfrac{256}{960}\rho L V^2 = \dfrac{4}{15}\rho L V^2$.

Step 3. Fraction in the fundamental

$\dfrac{E_1}{E} = \dfrac{256/\pi^6}{4/15} = \dfrac{960}{\pi^6} \approx 0.99856$: a smooth, symmetric strike puts 99.86% of the energy in the fundamental; the string sounds an almost pure tone of frequency $c/(2L)$.

Step 4. Periodicity

$\omega_n = n\omega_1$, so every term has period $2\pi/\omega_1 = 2L/c$ and $u(x, t + 2L/c) = u(x,t)$. Only odd n appear, and $\sin(\omega_n t + n\pi) = -\sin\omega_n t$ for odd n, hence $u(x, t + L/c) = -u(x,t)$. At $t = L/c$: $u = 0$ and $u_t = \sum \omega_n B_n \sin k_n x \cos n\pi = -g(x)$: the string crosses its rest position with the velocity reversed. At $t = L/(2c)$: $\cos(n\pi/2) = 0$ for odd n, so $u_t = 0$: the string is momentarily at rest at its maximum displacement (midpoint $VL/(3c)$).

KEY RESULTS

- $E_n = \dfrac{\rho L}{4}\omega_n^2 B_n^2 = \dfrac{256\,\rho L V^2}{\pi^6 n^6}$ (n odd), constant in time.
- $E(0) = \dfrac{4\rho L V^2}{15}$, and $\sum_{n\ \text{odd}} E_n = \dfrac{256\rho L V^2}{\pi^6}\cdot\dfrac{\pi^6}{960} = \dfrac{4\rho L V^2}{15}$.
- Fraction in the fundamental: $\dfrac{960}{\pi^6} \approx 0.99856$.
- Period $\dfrac{2L}{c}$; at $t = L/(2c)$ the string is momentarily at rest with maximum displacement; at $t = L/c$ it crosses the rest position with velocity $-g(x)$, and $u(x, t + L/c) = -u(x,t)$.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $c \to \infty$ (very stiff string): $u \to 0$, the displacement amplitude $VL/(3c)$ vanishes. Small t: $u \approx t g(x)$, consistent with the initial velocity. The coefficients decay as n^{-4}, faster than for a plucked string (n^{-2}), because the displacement is the time integral of a smooth velocity profile.
- **Checks:** dimensions: $B_n = \dfrac{32VL}{c\pi^4 n^4}$ is $\dfrac{(\mathrm{m/s})\,\mathrm{m}}{\mathrm{m/s}} = \mathrm{m}$; $E = \rho L V^2$ is (kg/m)(m)(m^2/s^2) = J. Two independent routes (series and d'Alembert) give the same midpoint displacement $VL/(3c)$.
- **Common mistakes in the exam:** (1) keeping $\cos\omega_n t$ terms although the initial displacement is zero; (2) setting B_n equal to the sine coefficients of g, forgetting the factor $1/\omega_n$ from differentiating in t; (3) using g itself instead of its odd periodic extension in d'Alembert's formula when $ct > L/2$; (4) giving the period as L/c: that is only the half-period after which u changes sign.